

On the Information Required for Feedback Control

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Biological systems across scales, along with many engineering problems, must control noisy systems with limited information. Here we study information-limited feedback control of stochastic systems to achieve target steady states, and derive a lower bound on the information rate from controlled system to controller. This framework allows us to obtain performance-information Pareto frontiers for wide-ranging control problems with limited information. For systems with state-independent passive dynamics, the bound is saturated by an explicit optimal control protocol which probabilistically time-reverses the passive dynamics. We showcase these results through applications to nonlinear particle localization, microbial navigation, and experimentally realized information engines.

I. INTRODUCTION

Control lies at the heart of many problems across biology, engineering, and beyond [1]. The problem of optimally controlling a physical system suffers from fundamental limitations including energetic costs, and limited information. Determining control protocols which minimize energy dissipation has been a central focus of stochastic thermodynamics [2–7]. Information-limited control has been less thoroughly explored, despite its importance in contexts such as single-cell decision-making [8], evolution [9], and engineering at the nanoscale [10].

A well-known result in this area is the Data Rate Theorem [11–13], a control-theoretic result which quantifies the minimum information rate required to stabilize an unstable linear dynamical system. Generalizations to nonlinear systems exist under certain conditions [14]. While a threshold for stabilization is useful, the problem of determining Pareto frontiers between information and performance remains unsolved. Related work [15–19] has explored information-limited control more generally, but has not yet led to widely applicable performance-information bounds for general control problems.

Information-limited control has also been studied in the context of information thermodynamics [20–22]. Previous work exploring the intersection of information thermodynamics and control theory has extended results from stochastic thermodynamics such as fluctuation theorems to feedback control [23–26], while the more recent thermodynamics of bipartite systems [27–32] has led to a deeper understanding of connections between heat extraction, energetic costs, and information flows between subsystems. Crucially, work in this area has focused primarily on systems and controllers with a clear thermodynamic interpretation. This restriction significantly limits applicability, especially to complex biological systems.

In this Article, we derive an inequality, Eq. (12), bounding steady-state control performance by the rate of

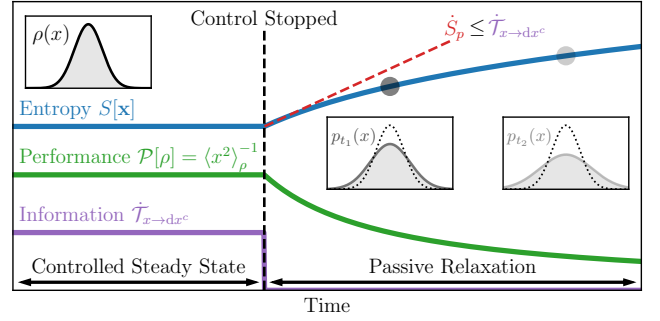


FIG. 1. Schematic illustrating the setup and our bound. A diffusive Brownian particle in one dimension (passive dynamics $dx_t^p = \sqrt{2D} dW_t^p$) is controlled to remain near to $x = 0$, producing a narrow steady-state distribution $\rho(x)$ and high performance $\mathcal{P}[\rho] = 1/\langle x^2 \rangle_\rho$. If the control is stopped at time t_0 , the system relaxes under the passive dynamics, with decreasing performance and increasing entropy $S[x]$. Our main result shows that the rate at which the entropy increases at the instant the control is stopped, $\dot{S}_p$, lower-bounds the information rate $\dot{\mathcal{T}}_{x \rightarrow dx^c}$ required for steady-state control.

information flow from the state of the system to the controller. This result shows explicitly that performance is always limited by information, and constitutes a powerful tool for deriving exact Pareto frontiers and scaling relationships in wide-ranging applications. For a broad class of systems, we derive an optimal control protocol which saturates the bound by probabilistically time-reversing the passive dynamics of the uncontrolled system.

We demonstrate the power of our results by studying several examples of increasing complexity. We first consider two analytically tractable model systems: controlling the state of a bit and localizing a diffusive particle in a nonlinear potential energy landscape. We then consider information-limited microbial navigation, deriving an explicit Pareto frontier and the optimal control protocol while extending recent work [8, 19] to control protocols with memory. Our framework allows us to prove that a memoryless Run-Reverse strategy is optimal in the low-information limit, suggesting an information-theoretic explanation for the behavior of real microbes.

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Finally, we consider the information cost of operating information engines, which lie at the heart of the intersection between information thermodynamics and control theory [33, 34]. Our bound implies a diverging information cost to approach the finite maximum rate of energy extraction from a thermal bath using feedback, which we verify in simulations of an experimentally realized information engine.

II. RESULTS

Consider a system with vector state $\mathbf{x}$. In the absence of control, it follows Markovian dynamics that we refer to as “passive”, indicated by superscript p . The state evolves according to the following stochastic differential equation:

$$d\mathbf{x}_t^p = \underbrace{\mathbf{F}(\mathbf{x})dt}_{\text{drift}} + \underbrace{\sqrt{2\mathbf{D}}d\mathbf{W}_t^p}_{\text{diffusion}} + \underbrace{\Delta\mathbf{x}dJ(\lambda(\mathbf{x}, \Delta\mathbf{x}))}_{\text{jumps}}. \quad (1)$$

Here $\mathbf{F}(\mathbf{x})$ is the drift force, $\mathbf{D}$ is the diffusion tensor, and $\mathbf{W}_t^p$ denotes a standard Wiener process. The last term represents discrete jumps of any distance $\Delta\mathbf{x}$, governed by a Poisson process dJ with rate $\lambda(\mathbf{x}, \Delta\mathbf{x})$. We assume the state space has no reflecting boundaries. The passive dynamics can equivalently be described by a Fokker–Planck equation for the time-dependent probability $p_t(\mathbf{x})$,

$$\begin{aligned} \frac{\partial}{\partial t}p_t(\mathbf{x}) = & -\nabla \cdot [\mathbf{F}(\mathbf{x})p_t(\mathbf{x})] - \int d\Delta\mathbf{x} \lambda(\mathbf{x}, \Delta\mathbf{x})p_t(\mathbf{x}) \\ & + \int d\Delta\mathbf{x} \lambda(\mathbf{x}-\Delta\mathbf{x}, \Delta\mathbf{x})p_t(\mathbf{x}-\Delta\mathbf{x}) + \nabla \cdot \mathbf{D}\nabla p_t(\mathbf{x}). \end{aligned} \quad (2)$$

We seek to control this system, forcing the state $\mathbf{x}$ to obey a steady-state distribution $\rho(\mathbf{x})$. The true goal will generally not be to control the full distribution $\rho(\mathbf{x})$, but rather a functional $\mathcal{P}[\rho]$ which quantifies performance. To achieve this control, a control update $d\mathbf{x}^c$ is added to the passive update $d\mathbf{x}^p$, such that their sum gives the total state update,

$$d\mathbf{x}_t = d\mathbf{x}_t^p + d\mathbf{x}_t^c(\{\mathbf{x}\}_{-\infty}^t, \{\mathbf{x}^c\}_{-\infty}^{t-dt}). \quad (3)$$

Here $d\mathbf{x}_t^c$ may depend on the entire history of the controlled system, $\{\mathbf{x}\}_{-\infty}^t$, and control updates $\{\mathbf{x}^c\}_{-\infty}^{t-dt}$.

Our goal is to bound the information required to achieve this control, which we quantify using the transfer entropy rate [35] from state $\mathbf{x}$ to control action $d\mathbf{x}^c$:

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow d\mathbf{x}^c} \equiv \lim_{dt \rightarrow 0} \frac{I[d\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{x}^c\}_{-\infty}^{t-dt}]}{dt}. \quad (4)$$

The transfer entropy rate quantifies the causal flow of information, accounting for memory via dependence on the past trajectories of both the system and control action.

A. Information Limit for Feedback Control

We will show that the information rate required for control is bounded below by the rate $\dot{S}_p$ (illustrated in Fig. 1) at which the passive dynamics increase the system Shannon entropy, $S[\mathbf{x}] \equiv \langle -\ln \rho(\mathbf{x}) \rangle_\rho$,

$$\begin{aligned} \dot{S}_p & \equiv \left. \frac{d}{dt} S[\mathbf{x}] \right|_{d\mathbf{x}^c=0} \\ & = \left\langle (\nabla \ln \rho(\mathbf{x})) \cdot \mathbf{D} \nabla \ln \rho(\mathbf{x}) \right\rangle_\rho + \langle \nabla \cdot \mathbf{F}(\mathbf{x}) \rangle_\rho \\ & \quad + \left\langle \int d\Delta\mathbf{x} \lambda(\mathbf{x}, \Delta\mathbf{x}) \ln \left[\frac{\rho(\mathbf{x})}{\rho(\mathbf{x} + \Delta\mathbf{x})} \right] \right\rangle_\rho. \end{aligned} \quad (5)$$

The three terms respectively quantify contributions from diffusion, drift, and jumps in the passive dynamics (1). Here and throughout, unless otherwise specified, angle brackets denote an average over the distribution $\rho(\mathbf{x})$.

To derive our main result, we consider the control and passive updates of the state applied in sequence:

$$\mathbf{x}_t \xrightarrow{+d\mathbf{x}_t^c} \mathbf{y}_t \xrightarrow{+d\mathbf{x}_t^p} \mathbf{x}_{t+dt} \quad (6)$$

Here we have chosen an ordering of the two updates, but the result can also be derived with the opposite ordering, and in any case the ordering does not matter in the $dt \rightarrow 0$ limit. The intermediate state $\mathbf{y}_t \equiv \mathbf{x}_t + d\mathbf{x}_t^c$ may depend on the full history of the system $\{\mathbf{x}\}_{-\infty}^t$ as well as the control history $\{\mathbf{x}^c\}_{-\infty}^t$. The passive update $d\mathbf{x}_t^p$, by contrast, depends only on $\mathbf{y}_t$ since the passive dynamics are Markovian.

Now consider the increase in entropy due to the passive dynamics, $\Delta S_p \equiv S[\mathbf{x}_{t+dt}] - S[\mathbf{y}_t]$. Assuming stationarity, this must equal the entropy decrease due to active control, $\Delta S_p = S[\mathbf{x}_t] - S[\mathbf{y}_t]$. Note next that the marginal entropies of $\mathbf{x}_t$ and $\mathbf{y}_t$ can be expanded as

$$S[\mathbf{x}_t] = S[\mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^{t-dt}] + I[\mathbf{x}_t; \{\mathbf{x}^c\}_{-\infty}^{t-dt}], \quad (7)$$

and

$$\begin{aligned} S[\mathbf{y}_t] & = S[\mathbf{y}_t | \{\mathbf{x}^c\}_{-\infty}^t] + I[\mathbf{y}_t; \{\mathbf{x}^c\}_{-\infty}^t] \\ & = S[\mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^t] + I[\mathbf{y}_t; \{\mathbf{x}^c\}_{-\infty}^t]. \end{aligned} \quad (8)$$

In the second line, we used the fact that $\mathbf{y}_t$ is a deterministic translation of $\mathbf{x}_t$ given $d\mathbf{x}_t^c$. We then expand the passive entropy change using Eqs. (7) and (8) to obtain

$$\begin{aligned} \Delta S_p & = S[\mathbf{x}_t] - S[\mathbf{y}_t] \\ & = I[d\mathbf{x}_t^c; \mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^{t-dt}] \\ & \quad + I[\mathbf{x}_t; \{\mathbf{x}^c\}_{-\infty}^{t-dt}] - I[\mathbf{y}_t; \{\mathbf{x}^c\}_{-\infty}^t]. \end{aligned} \quad (9)$$

In the second line, we identified the conditional mutual information $I[d\mathbf{x}_t^c; \mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^{t-dt}]$ as the difference in conditional entropies $S[\mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^{t-dt}] - S[\mathbf{x}_t | \{\mathbf{x}^c\}_{-\infty}^t]$. Next, we note that the data-processing inequality [36] implies $I[\mathbf{y}_t; \{\mathbf{x}^c\}_{-\infty}^t] \geq I[\mathbf{x}_{t+dt}; \{\mathbf{x}^c\}_{-\infty}^t]$, and that

$I[\mathbf{d}\mathbf{x}_t^c; \mathbf{x}_t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] \leq I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]$ since $\{\mathbf{x}\}_{-\infty}^t$ cannot be less informative about $\mathbf{d}\mathbf{x}_t^c$ than $\mathbf{x}_t$ alone. Together, these give

$$\Delta S_p \leq I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] + I[\mathbf{x}_t; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] - I[\mathbf{x}_{t+dt}; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^t]. \quad (10)$$

Since the system is at steady state, the terms in the second line are equal so that their difference is zero, leaving

$$\Delta S_p \leq I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]. \quad (11)$$

Dividing by dt and taking the limit $dt \rightarrow 0$ leads to our first main theoretical result:

$$\dot{S}_p \leq \dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}. \quad (12)$$

Intuitively, the information required to control a system must at least match the rate at which the passive dynamics grow the volume in state space. This result is conceptually related to prior work [15, 16], but constitutes a significant advance: our bound relates the instantaneous passive entropy rate (rather than path-dependent entropies) to the control transfer entropy rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$, which quantifies causal information flow from system to controller.

The bound (12) enables the derivation of performance-information Pareto frontiers. The performance metric of interest generally depends only on the steady-state distribution $\rho(\mathbf{x})$. The information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$, meanwhile, depends on the full history of the dynamics. This non-locality makes it generally difficult to calculate, though recent progress has enabled tractable numerical computation [37, 38]. Our lower bound, the instantaneous passive entropy rate $\dot{S}_p$, similarly depends only on $\rho(\mathbf{x})$, and can thus more easily be related to the performance $\mathcal{P}[\rho]$.

Indeed, the problem of determining the steady-state distribution $\rho(\mathbf{x})$ which maximizes the performance $\mathcal{P}[\rho]$ at fixed passive entropy rate $\dot{S}_p$ can be mapped onto the problem of finding the ground state of a corresponding Schrödinger equation (App. B). This mapping facilitates analytic or numerical calculations of both Pareto frontiers and optimal controlled steady-state distributions. Once the optimal steady-state $\rho(\mathbf{x})$ has been determined, the remaining task is to find the control protocol which minimizes the information rate. We will consider this problem in the next section.

The bound (12) is most useful when the objective is to tightly localize the steady-state probability distribution, as in the examples considered in this Article. If instead the objective is to achieve a wider distribution than the passive dynamics would otherwise produce, $\dot{S}_p$ can be negative. We explore such a setup in App. D.

While the transfer entropy rate is the natural way to quantify information for a controller with access to memory, a more tractable information metric for memoryless control (where $\mathbf{d}\mathbf{x}_t^c$ depends only on $\mathbf{x}_t$) is the instantaneous information rate between state and control up-

date [19],

$$\dot{I}_c^+ \equiv \lim_{dt \rightarrow 0} \frac{I[\mathbf{x}_t; \mathbf{d}\mathbf{x}_t^c]}{dt}. \quad (13)$$

A closely related quantity was recently shown to constitute a lower bound on the entropy production rate [39]. This instantaneous information rate is generally much easier to compute, and upper-bounds the transfer entropy rate for Markovian control, so $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} \leq \dot{I}_c^+$ (see App. A). Computing the information-optimal control using $\dot{I}_c^+$ can in many cases yield a near-optimal strategy (e.g., Fig. 2).

B. Optimal and Near-Optimal Control by Time-Reversing the Passive Dynamics

To derive optimal control protocols which maximize performance with limited information, consider the gap in the bound (12), which is (App. C):

$$\begin{aligned} \dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} - \dot{S}_p &= \lim_{dt \rightarrow 0} \frac{1}{dt} I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^{t-dt} | \mathbf{x}_t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] \\ &\quad + \lim_{dt \rightarrow 0} \frac{1}{dt} I[\mathbf{y}_t; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^t | \mathbf{x}_{t+dt}]. \end{aligned} \quad (14)$$

Saturating the bound thus requires setting the two terms on the right-hand side to zero. The first term quantifies the information about the next control action contained in the history of the state, after conditioning on the current state and control history. This is zero if the control update $\mathbf{d}\mathbf{x}_t^c$ depends only on the current state $\mathbf{x}_t$. More generally, optimal protocols should carry no additional information about past history after conditioning on the current state and control history. The second term is less intuitive, quantifying the information the post-control state $\mathbf{y}_t$ carries about the control history after conditioning on the state $\mathbf{x}_{t+dt}$ after the passive update (as defined in Eq. (6)). Determining conditions which set this term to zero in general is challenging, but for a special class of systems this allows us to construct explicit optimal protocols.

Consider a restricted class of systems where the passive dynamics $\mathbf{d}\mathbf{x}_t^p$ do not depend on the state $\mathbf{x}_t$, such that

$$\mathbf{d}\mathbf{x}_t^p = \mathbf{F}dt + \sqrt{2\mathbf{D}}d\mathbf{W}_t^p + \Delta\mathbf{x}dJ(\lambda(\Delta\mathbf{x})), \quad (15)$$

for constant $\mathbf{F}$ and rate $\lambda(\Delta\mathbf{x})$ independent of $\mathbf{x}$. For such passive dynamics, optimal control is achieved by a control protocol which probabilistically time-reverses the passive kernel $K_p(\mathbf{x}'|\mathbf{x}) = p(\mathbf{x}_t + \mathbf{d}\mathbf{x}_t^p|\mathbf{x}_t)$, with

$$p(\mathbf{y}_t|\mathbf{x}_t) = \frac{K_p(\mathbf{x}_t|\mathbf{y}_t)p(\mathbf{y}_t)}{p(\mathbf{x}_t)}. \quad (16)$$

Evaluating the protocol for the passive dynamics in Eq. (15), we obtain an explicit optimal protocol:

$$\begin{aligned} \mathbf{d}\mathbf{x}_t^c &= [-\mathbf{F} + 2\mathbf{D}\nabla \ln \rho(\mathbf{x}_t)]dt + \sqrt{2\mathbf{D}}d\mathbf{W}_t^c \\ &\quad + \Delta\mathbf{x}dJ[\lambda(-\Delta\mathbf{x})\rho(\mathbf{x}_t + \Delta\mathbf{x})/\rho(\mathbf{x}_t)]. \end{aligned} \quad (17)$$

The first two terms probabilistically time-reverse the passive drift-diffusion, while the third addresses the passive jumps. This is our second main theoretical result, providing an explicit optimal protocol for information-limited control when the passive dynamics are state-independent. Time-reversal protocols have been shown previously to minimize entropy production for thermodynamically consistent control of information engines [40, 41]; here we show that probabilistic time reversal is the optimal protocol for a broad class of information-limited control problems regardless of thermodynamic considerations.

When the passive dynamics do depend on the state, finding an optimal protocol is less straightforward outside of special cases such as linear dynamics (App. D). In the high-information limit when the steady-state distribution is highly localized, however, time-reversal protocols should still be optimal since the passive dynamics will be approximately state-independent in the narrow domain where steady-state probability is high. Based on this argument, along with exploration of the example systems considered here, we conjecture that probabilistically time-reversing the passive dynamics constitutes a near-optimal control protocol more generally. In App. C we derive an explicit upper bound [Eq. (C17)] on the gap between the information rate of the time-reversal protocol and the lower bound (12) given by the passive entropy rate. This makes it straightforward to quantify the distance from optimality of the time-reversal protocol.

III. APPLICATIONS

A. Controlling the State of a Bit

To provide intuition for the time-reversal control protocol, consider the problem of controlling a single bit (with $\mathbf{x} \in \{0, 1\}$), which passively flips its state at rate γ [Fig. 2]. Here the goal of the controller is to maximize the performance metric $\mathcal{P}[\rho] = \rho_0$, the probability of the bit being in state 0. Applying the bound (12), the passive entropy rate gives a lower bound on the information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$ required for control:

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} \geq \dot{S}_p = (2\rho_0 - 1)\gamma \ln \frac{\rho_0}{1 - \rho_0}. \quad (18)$$

Given γ and a target performance ρ_0 , the controller can choose the control jump rates λ_0 and λ_1 to minimize the information rate. With both γ and ρ_0 fixed, the control protocol can be fully characterized by a single parameter, which we take to be the rate λ_0 at which the controller induces bit errors. One might expect the optimal controller to make no errors ($\lambda_0 = 0$), acting only when necessary to flip the bit from the incorrect state $x = 1$ at rate λ_1 . Computing the information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$, however, reveals a minimum at a finite error rate $\lambda_0 = \gamma(1 - \rho_0)/\rho_0$ as predicted by Eq. (17). This is precisely the rate required to ensure that the control dynamics probabilistically time-reverse the passive dynamics, given the target ρ_0 . The

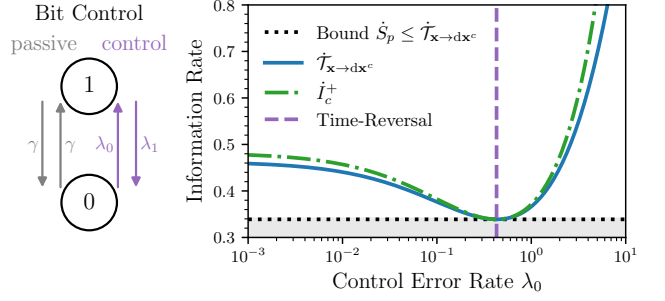


FIG. 2. Controlling a noisy bit. Left: schematic, with passive bit flips at rate $\gamma = 1$. The performance measure is $\mathcal{P}[\rho] = \rho_0$, the steady-state probability of being in state 0. Right: information rate vs control error rate λ_0 , always achieving a fixed performance $\rho_0 = 0.7$. Given ρ_0 and γ , λ_0 uniquely fixes λ_1 . Black dotted line: the bound Eq. (12). Blue line: information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$. Green dot-dashed line: instantaneous information rate $\dot{I}_c^+ \geq \dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$, Eq. (13). Purple dashed line: optimal rate λ_0 which saturates the bound by time-reversing the passive dynamics.

full-history transfer entropy rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$ requires a fairly involved calculation (App. E) even for this simple model system. The instantaneous information rate $\dot{I}_c^+$ (13), by contrast, is straightforward to compute analytically, and is also minimized by the probabilistic time-reversal protocol.

B. Localization in a Potential

As a more involved but still analytically tractable example, consider the problem of localizing a Brownian particle in a potential energy landscape $V(\mathbf{x})$. The passive dynamics (in scaled units) are

$$d\mathbf{x}_t^p = -\nabla V(\mathbf{x})dt + \sqrt{2}d\mathbf{W}_t^p. \quad (19)$$

For a given steady state $\rho(\mathbf{x})$, the passive entropy rate is

$$\dot{S}_p = \mathcal{I}[\rho] - \langle \nabla^2 V(\mathbf{x}) \rangle_\rho, \quad (20)$$

where $\mathcal{I}[\rho]$ is the Fisher information with respect to position shifts,

$$\mathcal{I}[\rho] \equiv \langle (\nabla \ln \rho(\mathbf{x})) \cdot \nabla \ln \rho(\mathbf{x}) \rangle_\rho. \quad (21)$$

Suppose the goal is to tightly localize the system near some desired point $\mathbf{x}^*$, maximizing performance $\mathcal{P}[\rho] = 1/\langle |\mathbf{x} - \mathbf{x}^*|^2 \rangle_\rho$. When the particle is tightly localized, approximating the potential energy landscape as locally quadratic using the Hessian $H_* = \nabla^2 V(\mathbf{x}^*)$ gives

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} \geq \dot{S}_p = \sum_{\lambda_i} \text{Re}(\lambda_i) + \mathcal{I}[\rho], \quad (22)$$

where $\{\lambda_i\}$ are the eigenvalues of $\mathbf{A} = -H_*$.

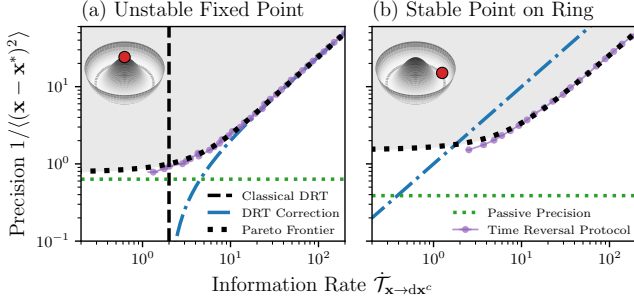


FIG. 3. Controlling a Brownian particle in a Mexican-hat potential, with the target being either (a) the unstable fixed point $\mathbf{x}_1^* = (0, 0)$, or (b) the stable fixed point $\mathbf{x}_2^* = (1, 0)$. With no control, the performance $\mathcal{P}[\rho] = 1/\langle |\mathbf{x} - \mathbf{x}^*|^2 \rangle$ remains finite (green dotted lines). The Data Rate Theorem (DRT), Eq. (23), provides a lower bound on the information rate required for stabilization (vertical dashed line) only in the unstable case (a). Blue dot-dashed curves: the DRT generalized to account for diffusion, Eq. (24). Black dotted curves: Pareto frontier from Eq. (12), with the forbidden region shaded in grey. Purple points: time-reversal protocol, Eq. (C16), with information rate computed numerically using the TE-PWS algorithm [38].

The form above is reminiscent of the Data Rate Theorem (DRT) [11], which states that to control an unstable linear system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, the minimum information required is the sum of the real parts of the eigenvalues of $\mathbf{A}$ with positive real part [11–13]:

$$\dot{I} \geq \sum_{\text{Re}(\lambda_i) > 0} \text{Re}(\lambda_i). \quad (23)$$

In the linear limit, we can connect our bound (12) to the DRT by first projecting the state $\mathbf{x}$ onto only the unstable directions to obtain a projected state $\mathbf{z}$, then restricting the controller to only act on $\mathbf{z}$ with control update that likewise depends only on $\{\mathbf{z}\}_{-\infty}^t$. Doing so yields a generalization of the DRT which quantifies the added information cost to counteract passive diffusion when controlling a passively unstable linear system:

$$\dot{T}_{\mathbf{z} \rightarrow d\mathbf{z}^c} \geq \sum_{\text{Re}(\lambda_i) > 0} \text{Re}(\lambda_i) + \mathcal{I}[\rho(\mathbf{z})]. \quad (24)$$

As an illustrative example, consider the two-dimensional Mexican-hat potential (in scaled units),

$$V(r) = -\frac{1}{2}r^2 + \frac{1}{4}r^4, \quad r = \sqrt{x^2 + y^2}. \quad (25)$$

The origin $r = 0$ is an unstable fixed point, while the ring with $r = 1$ is a set of stable fixed points. We consider the control problem of minimizing the mean-squared distance from the fixed points $\mathbf{x}_1^* = (0, 0)$, and $\mathbf{x}_2^* = (1, 0)$.

We begin with the target $\mathbf{x}_1^*$, which is an unstable fixed point. Linearized around $\mathbf{x}_1^*$, both eigenvalues are unstable and equal to 1, so the classical DRT gives $\dot{I} \geq 2$ as the

minimum information rate required to achieve stable control. The Cramér–Rao inequality implies $\mathcal{I}[\rho] \geq 4/\langle r^2 \rangle_\rho$, so that our bound (12) gives:

$$\dot{T}_{\mathbf{x} \rightarrow d\mathbf{x}^c} \geq 2 - 4\langle r^2 \rangle_\rho + 4/\langle r^2 \rangle_\rho. \quad (26)$$

While the Cramér–Rao bound is valid for arbitrary distributions, we show that it is saturated by the distribution ρ that maximizes $\mathcal{P}$ at fixed $\dot{S}_p$ (App. F). Thus, Eq. (26) is the Pareto frontier obtained from our main bound. For these optimal distributions, the time-reversal protocol (C16) is near-optimal [Fig. 3(a)]. Crucially, while the DRT provides a threshold for the information rate required to stabilize an unstable system, we instead obtain a full Pareto frontier valid even below the classical threshold of stability [Fig. 3(a)]. Since the Mexican-hat potential is ultimately confining for large $|\mathbf{x}|$, $\langle |\mathbf{x} - \mathbf{x}^*|^2 \rangle_\rho$ remains finite under the passive dynamics, even with no control.

We then consider the second target $\mathbf{x}_2^* = (1, 0)$ [Fig. 3(b)]. This point is stable under deterministic dynamics, so the DRT bound is simply 0. Eq. (12), by contrast, provides a full Pareto frontier for the problem which the time-reversal protocol saturates at high information. Thus our framework gives information limits for control even in cases where the DRT is uninformative. Appendix F provides full details of the calculations.

C. Navigation up a Gradient

Consider now a biological example, navigation up a gradient in two dimensions, motivated by microbial chemotaxis and explored in recent work [19]. We model chemotaxis as constant-speed motion (with speed v_0) at a heading angle $\theta \in [0, 2\pi)$ with respect to the gradient direction. The angle θ is both the signal acquired by the sensor and the variable on which the controller acts, obeying overdamped Langevin dynamics on the circle:

$$d\theta_t = \sqrt{2D_r} dW_t^p + d\theta_t^c. \quad (27)$$

The passive dynamics are rotational diffusion with coefficient D_r , such that the passive entropy rate is $\dot{S}_p = D_r \mathcal{I}[\rho(\theta)]$. The performance metric, meanwhile, is the up-gradient velocity, $\mathcal{P}[\rho] = v/v_0 = \langle \cos \theta \rangle_\rho$. To derive the velocity–information Pareto frontier, we maximize $\mathcal{P}[\rho]$ given $\dot{S}_p$ to obtain (App. G)

$$v/v_0 \leq \frac{1}{2} a'_0 \left[b_0^{-1} \left(\dot{T}_{\theta \rightarrow d\theta^c} / D_r \right) \right], \quad (28)$$

where $a_0(x)$ is the Mathieu characteristic function (and $a'_0(x)$ its derivative), and $b_0(x) = a_0(x) - x a'_0(x)$ [Fig. 4]. This shows that the velocity–information Pareto frontier obtained in our previous work [19] applies to the full history-dependent transfer entropy rate, thus generalizing to arbitrary control protocols with memory. At low information this frontier scales as $v/v_0 \sim \sqrt{\dot{T}_{\theta \rightarrow d\theta^c} / 2D_r}$, and at high information scales as $v/v_0 \sim 1 - D_r / 2\dot{T}_{\theta \rightarrow d\theta^c}$.

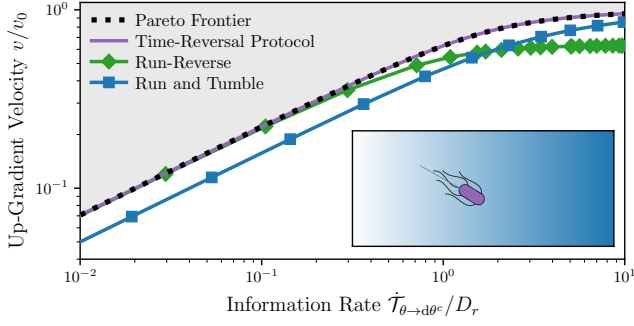


FIG. 4. Up-gradient velocity vs information rate for navigation in 2D (inset: schematic). Black dotted line: Pareto frontier, Eq. (28). Purple: time-reversal protocol. Blue Squares: optimized Run-and-Tumble navigation. Green Diamonds: optimized Run-Reverse navigation, with information rate computed using TE-PWS [38].

The Pareto frontier is exactly saturated by the probabilistic time-reversal protocol (17), with no jumps, and drift force expressible as special functions of the information rate (App. G). This Markovian control strategy was derived previously by maximizing velocity with respect to the instantaneous control information rate $\dot{I}_c^+$ [19], but also maximizes velocity with respect to the full-history transfer entropy rate (in fact, this optimal protocol has $\dot{I}_c^+ = \dot{T}_{\theta \rightarrow d\theta^c}$). Thus memory of past states and actions cannot improve performance.

The optimal control protocol described above requires full control over a steering force at every possible heading angle θ , which is beyond the capabilities of the microbes such as *E. coli* which motivated the setup. Such microbes typically measure the time derivative of the concentration (or its logarithm), roughly analogous to measuring $\cos \theta$ in our simplified model. Given this constraint, a common navigation strategy in the microbial world (employed by, e.g., *E. coli* [8]) is to Run-and-Tumble, where the only control action is to fully reorient the heading with rate $\lambda(\{\theta\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^t)$ such that the angle change $\Delta\theta$ is uniformly distributed on $[0, 2\pi]$. This strategy performs markedly worse than the optimal protocol [Fig. 4].

Given the observability constraint of measuring only $\cos \theta$, and allowing control actions $d\theta^c$ with arbitrary memory, we prove in App. G that in the low-information limit, the unique protocol which asymptotically saturates the Pareto frontier is an instantaneous Run-Reverse strategy that does not use memory of previous observations [Fig. 4]. This navigation strategy, implemented by engaging in turns of $\Delta\theta = \pi$ at a heading-dependent rate $\lambda(\theta)$, is employed by real-world bacteria such as *P. aeruginosa* and *V. cholerae* [42]. Given that *in vivo* information rates for bacterial chemotaxis have been estimated at $\mathcal{O}(10^{-2})$ bit/s [8], it is thus plausible that these microbes have evolved to navigate using the optimal control strategy given their limited sensory information and constrained observability. Beyond the low-information limit,

we expect a dauntingly complex set of distinct behavioral strategies will emerge, perhaps employing turns at discrete angles as found previously in Ref. [19], or perhaps leveraging memory to approximate the unconstrained optimal strategy. Any such strategies reliant on memory must still abide by the Pareto frontier (28).

D. Feedback Control Information Engine

Consider the colloidal information engine which has been implemented experimentally in Ref. [43]. The controlled system is a Brownian particle of mass m with vertical position x , diffusing in a harmonic potential with center z . The overdamped Langevin dynamics are

$$dx_t = \beta D \kappa (z_t - x_t) dt - \frac{mg}{\gamma} dt + \sqrt{2D} dW_t^p, \quad (29)$$

with control update $dz_t = dz_t^c$. Here D is the particle's diffusion coefficient, κ is the trap strength, g is the gravitational acceleration, and β is the inverse temperature. The controller moves the trap center z , and we require that the controller do no work on the particle on average, i.e., that the controller does not change the potential energy $V(x, z) = \frac{1}{2} \kappa (z - x)^2$. Performance is quantified by the rate, $\mathcal{P}[\rho] = \beta \dot{Q}$, at which heat is extracted from the bath, equal to the rate of free energy storage as gravitational potential energy.

To apply the bound (12), consider the difference coordinate $u = z - x$ which reaches a steady state $\rho(u)$. Defining the timescale $\tau_r = 1/\beta D \kappa$ and dimensionless mass $\delta_g = mg/\kappa \sqrt{D \tau_r}$, imposing the zero-average-work condition, and applying the Cramér-Rao inequality, we derive a Pareto frontier bounding the rate at which free energy can be extracted for a given information rate from state u to control action dz^c (App. H):

$$\tau_r \beta \dot{Q} \leq \frac{-\delta_g^2 + \delta_g \sqrt{\delta_g^2 + 4} / \left(1 + (\tau_r \dot{T}_{u \rightarrow dz^c})^{-1}\right)}{2}. \quad (30)$$

This upper bound increases monotonically with the information rate, asymptotically approaching the finite value $\frac{\delta_g}{2} \left(-\delta_g + \sqrt{\delta_g^2 + 4}\right)$ as $\dot{T}_{u \rightarrow dz^c} \rightarrow \infty$. Thus maximizing performance requires a diverging information rate.

The second law of information thermodynamics requires a minimum information rate of $\dot{T}_{u \rightarrow dz^c} \geq \beta \dot{Q}$ (App. H), leaving open the possibility that the maximum rate of heat extraction could be achieved using only a finite information rate. Our results, by contrast, show that the theoretical maximum performance can only be approached asymptotically with a diverging information rate. In the low-information limit the bound (30) reduces to $\beta \dot{Q} \leq \dot{T}_{u \rightarrow dz^c}$, recovering the second law.

In the experimental implementation of Ref. [43], the controller measures the particle position $\hat{x}_t = x_t + \mathcal{N}(0, \sigma_m^2)$ (with measurement noise σ_m) at periodic time

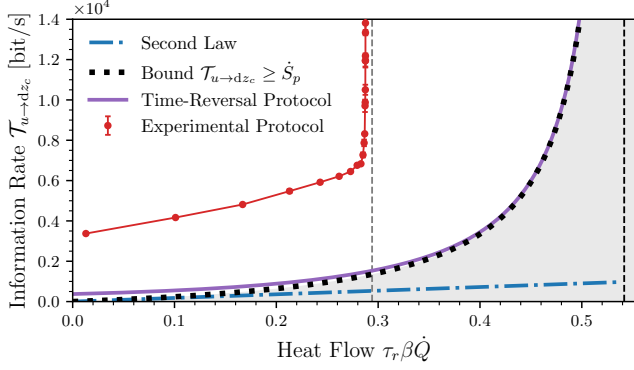


FIG. 5. Information rate vs heat extracted from the bath for an information engine. Blue: second law of information thermodynamics. Black dotted line: Pareto frontier (30). Red points: simulations of the experimentally implemented control protocol from Ref. [43] plotted parametrically, varying the measurement noise σ_m , with information rate evaluated using TE-PWS [38]. Purple line: Time-reversal protocol. Black dot-dashed line: second law of bipartite thermodynamics. Vertical dashed lines: maximum heat flow with no average work (black) and with no work for each controller action (grey). Parameters are $t_s/\tau_r = 1/41$, $\delta_g = 0.8$, and $\tau_r = 0.8 \times 10^{-3}$ s, from the experimental setup in Ref. [10].

intervals of length t_s , and moves the trap center z by a distance dz^c which depends on the controller's estimate of u_t . The control protocol updates the trap center based only on the measured particle position as

$$dz_{t=nt_s}^c = \begin{cases} \alpha(\hat{x}_t - z_t), & \hat{x}_t > z_t, \\ 0, & \text{otherwise,} \end{cases} \quad (31)$$

for integer n , and feedback gain parameter α that is tuned to ensure the controller does no average work. For simplicity, we omit the feedback delay in Ref. [43]; this would be a useful direction for future work.

The prediction of diverging information cost is confirmed in simulations for the experimentally implemented control protocol (Fig. 5). Because the protocol is restricted such that the only possible trap center movement is a reflection about the particle position x , the maximum free energy extraction is lower than predicted by Eq. (30). Relaxing this restriction and considering more general control strategies (allowing temporary energy exchange between the particle and the controller [44]) unlocks significant performance improvements. In App. H we construct a near-optimal control protocol to probabilistically time-reverse the passive dynamics, which dramatically outperforms the experimentally implemented protocol (Fig. 5).

IV. DISCUSSION

In this Article we derived a lower bound on the information rate required for feedback control of stochastic

systems. The bound, the passive entropy rate $\dot{S}_p$, requires only knowledge of the passive dynamics and target distribution $\rho(\mathbf{x})$, and can easily be further related to performance metrics of interest. This makes Eq. (12) a powerful tool for deriving information–performance Pareto frontiers for varied control problems.

Examining saturation conditions for the bound (12) leads to an explicit optimal control protocol (Eq. (17)) for state-independent passive dynamics, in which the controller probabilistically time-reverses the passive dynamics. More generally, we show that time-reversal protocols are near-optimal for wide-ranging applications, demonstrating the utility of this insight. Since optimal protocols are inherently probabilistic, they can in many cases be implemented even in the presence of noise arising from measurement, decision-making, or actuation. For real-world implementation, probabilistic protocols may be costly to implement on deterministic digital computers, but are ideally suited for devices such as thermodynamic computers [45, 46] or physical learning systems [47, 48].

While intimately related to the Data Rate Theorem [11–13], our results constitute a significant advancement, generalizing to arbitrary stochastic dynamics, providing full Pareto frontiers rather than just thresholds required for stabilization, and providing finite bounds even when the DRT suggests no information is required for control. In the application to microbial navigation, our bound leads to a tight velocity–information Pareto frontier, and proves that memory cannot improve control when the state is fully observable. Under biologically realistic observability constraints, our framework shows that in the low-information limit, the optimal navigation strategy is Run-Reverse with no memory, as implemented by real bacteria such as *P. aeruginosa*. For feedback-control information engines, our bound improves on the second law of information thermodynamics by deriving a diverging information cost to approach the theoretical maximum free energy extraction from a thermal bath.

Here we focused on feedback control to maintain target steady states. Time-dependent control constitutes another important class of problems in control theory, where the goal of a control protocol is to move the system from one state to another in finite time. Minimizing the first-passage time is also an important distinct objective relevant in many areas of biology. We suspect our framework can be generalized to such control protocols; this will be an important direction for future work. We also restricted our attention to the control of Markovian stochastic systems. Since non-Markovian systems can almost always be mapped onto Markovian dynamics with an expanded state space, we do not expect this to significantly limit the applicability of our results. Nonetheless, given the prevalence of long-range non-Markovian memory throughout biology [49], explicitly exploring the control of non-Markovian systems could prove insightful.

This work focused on the information cost of control, ignoring thermodynamic considerations such as energetic costs. Our approach, being agnostic to thermodynamic

considerations, is well suited for application to complex biological systems where the coupling between control and energetic costs is not easily identified. Nonetheless, energetic constraints are central to many control problems of interest, so understanding the relationship between information and energetic costs of control will be an important future direction. Perhaps the information rate $\dot{\mathcal{T}}_{x \rightarrow dx^c}$ lower-bounds the rate of energy consumed by the controller through a Landauer-like inequality, though we do not yet have a proof of this.

Finally, this Article necessarily considered only a handful of illustrative applications. We envision the present framework and bound for information-limited control being applicable to control problems across biology, chem-

istry, physics, engineering, and economics, and hope to see such applications explored in future work.

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Code Availability.—Python code for reproducing all figures in the manuscript is available on GitHub [50].

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Appendix A: Computing and Bounding Information Rates

Here we provide general details for computing and bounding the control transfer entropy rate, defined as

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} \equiv \lim_{dt \rightarrow 0} \frac{I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]}{dt}. \quad (\text{A1})$$

Evaluating the conditional mutual information gives

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} = \lim_{dt \rightarrow 0} \frac{1}{dt} \left\langle \left\langle \ln \frac{p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{x}\}_{-\infty}^t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})}{p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})} \right\rangle_{p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{x}\}_{-\infty}^t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})} \right\rangle_{p(\{\mathbf{x}\}_{-\infty}^t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})}. \quad (\text{A2})$$

Thus evaluating the information rate requires knowledge of the control update distribution given the full history of both the system and controller, $p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{x}\}_{-\infty}^t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})$, as well as the distribution of the control update given only the control history, $p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt})$. This second distribution is generally extremely difficult to compute outside of special cases.

To address the difficulty of computing the full-history transfer entropy rate, we show that the instantaneous control information rate,

$$\dot{I}_c^+ = \lim_{dt \rightarrow 0} \frac{I[\mathbf{x}_t; \mathbf{d}\mathbf{x}_t^c]}{dt}, \quad (\text{A3})$$

is an upper bound on $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c}$ for control dynamics which do not rely on memory. Without memory the control is Markovian, with $p(\mathbf{d}\mathbf{x}_t^c | \{\mathbf{x}\}_{-\infty}^t, \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}) = p(\mathbf{d}\mathbf{x}_t^c | \mathbf{x}_t)$. In this case the transfer entropy rate obeys

$$\begin{aligned} \dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} &= \lim_{dt \rightarrow 0} \frac{I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]}{dt} \\ &= \lim_{dt \rightarrow 0} \frac{I[\mathbf{d}\mathbf{x}_t^c; \mathbf{x}_t | \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]}{dt} \\ &= \lim_{dt \rightarrow 0} \frac{1}{dt} (I[\mathbf{d}\mathbf{x}_t^c; \mathbf{x}_t] - I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] + I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt} | \mathbf{x}_t]) \\ &= \dot{I}_c^+ - \lim_{dt \rightarrow 0} \frac{I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}]}{dt} + 0 \\ &\leq \dot{I}_c^+. \end{aligned} \quad (\text{A4})$$

This also illustrates that the instantaneous control information and transfer entropy rate are equal when $I[\mathbf{d}\mathbf{x}_t^c; \{\mathbf{d}\mathbf{x}^c\}_{-\infty}^{t-dt}] = 0$, i.e., the control history alone contains no information about the next control update.

If instead the control strategy is non-Markovian, then this inequality does not hold. If we expand the state space to include a memory variable M_t such that the control depends only on $\mathbf{x}_t$ and M_t , then the preceding argument implies

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{d}\mathbf{x}^c} \leq \lim_{dt \rightarrow 0} \frac{I[\mathbf{x}_t, M_t; \mathbf{d}\mathbf{x}_t^c]}{dt}. \quad (\text{A5})$$

The instantaneous control information rate (13) is particularly useful because the information rate $\dot{I}_c^+$ is generally straightforward to compute. Suppose the control takes the form

$$\mathbf{d}\mathbf{x}_t^c = \mathbf{F}_c(\mathbf{x}_t)dt + \sqrt{2\mathbf{D}_c} d\mathbf{W}_t^c + \Delta\mathbf{x} dJ(\lambda_c(\mathbf{x}, \Delta\mathbf{x})), \quad (\text{A6})$$

which we showed previously can accommodate the optimal probabilistic time-reversal protocol. Then the instantaneous control information rate can be evaluated as [19]

$$\dot{I}_c^+ = \frac{1}{4} \left[\langle \mathbf{F}_c \cdot \mathbf{D}_c^{-1} \mathbf{F}_c \rangle_\rho - \langle \mathbf{F}_c \rangle_\rho \cdot \mathbf{D}_c^{-1} \langle \mathbf{F}_c \rangle_\rho \right] + \iint d\mathbf{x} d\Delta\mathbf{x} \rho(\mathbf{x}) \lambda_c(\mathbf{x}, \Delta\mathbf{x}) \ln \frac{\lambda_c(\mathbf{x}, \Delta\mathbf{x})}{\bar{\lambda}_c(\Delta\mathbf{x})}, \quad (\text{A7})$$

where $\bar{\lambda}_c(\Delta\mathbf{x}) = \int d\mathbf{x} \rho(\mathbf{x}) \lambda_c(\mathbf{x}, \Delta\mathbf{x})$.

Appendix B: Deriving Pareto Frontiers

Here we develop a general procedure for deriving performance-information Pareto frontiers for information-limited control problems. We begin with the passive entropy rate, which we write as an integral over $\mathbf{x}$:

$$\dot{S}_p = \int d\mathbf{x} \rho(\mathbf{x}) \left[(\nabla \ln \rho(\mathbf{x})) \cdot \mathbf{D} \nabla \ln \rho(\mathbf{x}) + \nabla \cdot \mathbf{F}(\mathbf{x}) + \int d\Delta \mathbf{x} \lambda(\mathbf{x}, \Delta \mathbf{x}) \ln \frac{\rho(\mathbf{x})}{\rho(\mathbf{x} + \Delta \mathbf{x})} \right]. \quad (\text{B1})$$

We further assume the performance metric can be written as the average of some scalar function $A(\mathbf{x})$:

$$\mathcal{P}[\rho] = \int d\mathbf{x} \rho(\mathbf{x}) A(\mathbf{x}). \quad (\text{B2})$$

Since we have shown the bound (12) between passive entropy rate and information rate can be tight, the Pareto frontier for a given problem can be determined by extremizing $\mathcal{P}$ at fixed $\dot{S}_p$. This can be achieved using the objective function

$$\mathcal{L} = \mathcal{P}[\rho] - \gamma \dot{S}_p + \alpha \int d\mathbf{x} \rho(\mathbf{x}). \quad (\text{B3})$$

Here γ and α are Lagrange multipliers, respectively enforcing constant $\dot{S}_p$ and normalization of $\rho(\mathbf{x})$. The condition for extremizing $\mathcal{L}$ is derived by setting the functional derivative $\delta \mathcal{L} / \delta \rho = 0$.

In the absence of passive jumps, this is a standard variational calculus problem. Defining $\psi(\mathbf{x})$ such that $\psi^2(\mathbf{x}) = \rho(\mathbf{x})$, $\lambda \equiv 1/\gamma$, and $E_0 = \alpha/\gamma$, the Euler–Lagrange equation gives

$$[-4\nabla \cdot \mathbf{D} \nabla + \nabla \cdot \mathbf{F}(\mathbf{x}) - \lambda A(\mathbf{x})] \psi(\mathbf{x}) = E_0(\lambda) \psi(\mathbf{x}). \quad (\text{B4})$$

This can be mapped onto a Schrödinger equation by defining the potential $U_\lambda(\mathbf{x}) = \nabla \cdot \mathbf{F}(\mathbf{x}) - \lambda A(\mathbf{x})$. If the diffusion matrix is diagonal and isotropic ($\mathbf{D} = D\mathbf{I}$), then we have

$$\underbrace{[-4D\nabla^2 + U_\lambda(\mathbf{x})]}_{\equiv H_\lambda} \psi(\mathbf{x}) = E_0(\lambda) \psi(\mathbf{x}), \quad (\text{B5})$$

where we have defined the Hamiltonian H_λ . This is a linear, time-independent Schrödinger equation, and suggests identification of the normalization Lagrange multiplier E_0 with the ground-state energy of the Hamiltonian H_λ , while the square of the ground-state wavefunction $\psi_0(\mathbf{x})$ gives the probability distribution $\rho(\mathbf{x})$ which extremizes $\mathcal{L}$. The ground state is the solution which extremizes the objective function.

The mapping can still be performed if $\mathbf{D}$ is non-diagonal and/or anisotropic through a change of variables. Since the diffusion matrix $\mathbf{D}$ must be positive definite, so can always be scaled out by changing to a variable $\mathbf{y}$ satisfying $\mathbf{x} = [(\mathbf{D} + \mathbf{D}^\top)/2]^{1/2} \mathbf{y}$. This would then yield a linear and time-independent Schrödinger equation in the variable $\mathbf{y}$. Finally, in the case where passive jumps are present, an additional term will be added to the Hamiltonian H_λ , making the problem in general map to a nonlinear and nonlocal time-independent Schrödinger equation.

Since Eq. (B5) is a linear, time-independent Schrödinger equation, many analytic solutions exist and are well documented. Indeed, examples studied in this Article map onto quantum mechanics problems such as the two-state system (bit flip control), the quantum harmonic oscillator (Mexican-hat potential, by minimizing mean-squared distance instead of maximizing its inverse), and the quantum rotor [51] (microbial navigation), for which analytical solutions for the ground state and energy are well-known. Even when analytic solutions are not possible, the linear time-independent Schrödinger equation has been thoroughly studied in the literature, and numerical techniques for efficiently computing ground states and energies are abundant. Solutions to the nonlinear and nonlocal version are generally far less tractable, but numerical techniques nonetheless exist.

The ground-state energy $E_0(\lambda)$, once determined, fully characterizes the performance-information Pareto frontier. Taking the derivative of H_λ with respect to λ , multiplying by $\psi^*(\mathbf{x})$, and integrating over $\mathbf{x}$, we find that the performance is given by the derivative E'_0 with respect to the Lagrange multiplier λ ,

$$\mathcal{P}[\rho] = -E'_0(\lambda). \quad (\text{B6})$$

Moreover, the minimum information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow d\mathbf{x}^c}$, equal to the passive entropy rate $\dot{S}_p$ can then be written in terms of the ground-state energy as

$$\dot{\mathcal{T}}_{\mathbf{x} \rightarrow d\mathbf{x}^c} = E_0(\lambda) - \lambda E'_0(\lambda) \equiv B_0(\lambda), \quad (\text{B7})$$

where we have defined the function $B_0(\lambda)$ in terms of E_0 and E'_0 . The Pareto frontier between performance and information is then, in full generality,

$$\mathcal{P}[\rho] = -E'_0 \left(B_0^{-1} \left[\dot{\mathcal{T}}_{\mathbf{x} \rightarrow d\mathbf{x}^c} \right] \right). \quad (\text{B8})$$

Appendix C: Time-Reversal Control Protocol

Here we construct a control protocol which saturates our bound (12) in the case where the passive dynamics are state-independent, i.e., $d\mathbf{x}_t^p$ is independent of $\mathbf{x}_t$. We assume that the state $\mathbf{x}_t$ is fully observable, and that there are no restrictions on the actions available to the controller.

We begin by noting that two inequalities were used in deriving Eq. (12). The first was the data-processing inequality,

$$I[\mathbf{y}_t; \{d\mathbf{x}^c\}_{-\infty}^t] \geq I[\mathbf{x}_{t+dt}; \{d\mathbf{x}^c\}_{-\infty}^t]. \quad (\text{C1})$$

The second was

$$I[d\mathbf{x}_t^c; \mathbf{x}_t | \{d\mathbf{x}^c\}_{-\infty}^{t-dt}] \leq I[d\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^t | \{d\mathbf{x}^c\}_{-\infty}^{t-dt}]. \quad (\text{C2})$$

Accounting for the difference between the left- and right-hand sides of these two inequalities, we find that the difference between the transfer entropy and passive entropy increase satisfies the equality

$$\Delta\mathcal{T}_{\mathbf{x} \rightarrow d\mathbf{x}^c} - \Delta S_p = \underbrace{I[d\mathbf{x}_t^c; \{\mathbf{x}\}_{-\infty}^{t-dt} | \mathbf{x}_t, \{d\mathbf{x}^c\}_{-\infty}^{t-dt}]}_{\text{Term 1}} + \underbrace{I[\mathbf{y}_t; \{d\mathbf{x}^c\}_{-\infty}^t | \mathbf{x}_{t+dt}]}_{\text{Term 2}}. \quad (\text{C3})$$

Term 1 is equal to zero if and only if the control update does not depend on the history of $\mathbf{x}$ given the current state $\mathbf{x}_t$ and past control history. This condition is satisfied if

$$p(d\mathbf{x}_t^c | \mathbf{x}_t, \{\mathbf{x}\}_{-\infty}^{t-dt}, \{d\mathbf{x}^c\}_{-\infty}^{t-dt}) = p(d\mathbf{x}_t^c | \mathbf{x}_t, \{d\mathbf{x}^c\}_{-\infty}^{t-dt}). \quad (\text{C4})$$

Term 2 is zero if and only if the previous control target $\mathbf{y}_t$ is independent of the control history $\{d\mathbf{x}^c\}_{-\infty}^t$ given $\mathbf{x}_{t+dt}$. This is achieved if

$$p(\mathbf{y}_t | \mathbf{x}_{t+dt}, \{d\mathbf{x}^c\}_{-\infty}^t) = p(\mathbf{y}_t | \mathbf{x}_{t+dt}). \quad (\text{C5})$$

We now construct a control protocol that satisfies these two conditions. First note that, at steady state, $\mathbf{x}_t$ and $\mathbf{x}_{t+dt}$ both follow the distribution $\rho(\mathbf{x})$. The variable $\mathbf{y}_t$, meanwhile, follows a (generally different) distribution $\rho'(\mathbf{y})$. Defining the passive transition kernel

$$K_p(\mathbf{x}' | \mathbf{y}) \equiv p(\mathbf{x}_{t+dt} | \mathbf{y}_t), \quad (\text{C6})$$

these two distributions are related as

$$\rho(\mathbf{x}') = \int d\mathbf{y} K_p(\mathbf{x}' | \mathbf{y}) \rho'(\mathbf{y}). \quad (\text{C7})$$

Now define the reverse transition kernel,

$$R_p(\mathbf{y} | \mathbf{x}') \equiv p(\mathbf{y}_t | \mathbf{x}_{t+dt}). \quad (\text{C8})$$

By Bayes' rule, this must satisfy

$$R_p(\mathbf{y} | \mathbf{x}') = \frac{\rho'(\mathbf{y}) K_p(\mathbf{x}' | \mathbf{y})}{\rho(\mathbf{x}')} \quad (\text{C9})$$

Finally, define the active transition kernel as

$$K_c(\mathbf{y} | \mathbf{x}) \equiv p(\mathbf{y}_t | \mathbf{x}_t), \quad (\text{C10})$$

and set it equal to the passive reverse kernel

$$K_c(\mathbf{y} | \mathbf{x}) = R_p(\mathbf{y} | \mathbf{x}). \quad (\text{C11})$$

This protocol depends only on the current state $\mathbf{x}_t$, so sets Term 1 to zero.

To see that Term 2 is also set to zero by this protocol, note that for state-independent passive dynamics the passive kernel takes the form $K_p(\mathbf{x}' | \mathbf{y}) = q(\mathbf{x}' - \mathbf{y})$ for some function q . To do so, we will first show that the distribution

$p(\mathbf{x}_{t+dt}|\{\mathbf{dx}^c\}_{-\infty}^t) = \rho(\mathbf{x}_{t+dt})$ does not depend on the control history, doing so by induction. Let us suppose first that $p(\mathbf{x}_t|\{\mathbf{dx}^c\}_{-\infty}^{t-dt}) = \rho(\mathbf{x}_t)$. We then compute

$$\begin{aligned} p(\mathbf{y}_t, \mathbf{x}_{t+dt}, \mathbf{dx}_t^c|\{\mathbf{dx}^c\}_{-\infty}^{t-dt}) &= p(\mathbf{x}_t = \mathbf{y}_t - \mathbf{dx}_t^c|\{\mathbf{dx}^c\}_{-\infty}^{t-dt}) \cdot K_c(\mathbf{y}_t|\mathbf{x}_t) \cdot K_p(\mathbf{x}_{t+dt}|\mathbf{y}_t) \\ &= \rho(\mathbf{y}_t - \mathbf{dx}_t^c) \cdot \frac{\rho'(\mathbf{y}_t)q(\mathbf{x}_t - \mathbf{y}_t)}{\rho(\mathbf{x}_t)} \cdot q(\mathbf{x}_{t+dt} - \mathbf{y}_t) \\ &= \rho'(\mathbf{y}_t)q(-\mathbf{dx}_t^c)q(\mathbf{x}_{t+dt} - \mathbf{y}_t). \end{aligned} \quad (\text{C12})$$

Integrating over $\mathbf{y}_t$ and $\mathbf{x}_{t+dt}$ then gives $p(\mathbf{dx}_t^c|\{\mathbf{dx}^c\}_{-\infty}^{t-dt}) = q(-\mathbf{dx}_t^c)$. We can then write

$$p(\mathbf{y}_t, \mathbf{x}_{t+dt}|\{\mathbf{dx}^c\}_{-\infty}^t) = \frac{p(\mathbf{y}_t, \mathbf{x}_{t+dt}, \mathbf{dx}_t^c|\{\mathbf{dx}^c\}_{-\infty}^{t-dt})}{p(\mathbf{dx}_t^c|\{\mathbf{dx}^c\}_{-\infty}^{t-dt})} = \rho'(\mathbf{y}_t)q(\mathbf{x}_{t+dt} - \mathbf{y}_t). \quad (\text{C13})$$

Integrating over $\mathbf{y}_t$ gives $p(\mathbf{x}_{t+dt}|\{\mathbf{dx}^c\}_{-\infty}^t) = \rho(\mathbf{x}_{t+dt})$, thus proving that $\mathbf{x}_t$ being independent of $\{\mathbf{dx}^c\}_{-\infty}^t$ is self-consistent and true by induction. Finally, computing

$$p(\mathbf{y}_t|\mathbf{x}_{t+dt}, \{\mathbf{dx}^c\}_{-\infty}^t) = \frac{p(\mathbf{y}_t, \mathbf{x}_{t+dt}|\{\mathbf{dx}^c\}_{-\infty}^t)}{p(\mathbf{x}_{t+dt}|\{\mathbf{dx}^c\}_{-\infty}^t)} = \frac{\rho'(\mathbf{y}_t)q(\mathbf{x}_{t+dt} - \mathbf{y}_t)}{\rho(\mathbf{x}_{t+dt})} = R_p(\mathbf{y}_t|\mathbf{x}_{t+dt}) \quad (\text{C14})$$

shows that Eq. (C5) is satisfied, and thus Term 2 is zero.

Thus an optimal control protocol for state-independent passive dynamics is the Bayesian time reversal of the passive dynamics. The protocol samples the post-control state $\mathbf{y}$ from the posterior distribution of where the passive dynamics must have come from to arrive at the pre-control state $\mathbf{x}_t$. We can evaluate the control dynamics explicitly given the Markovian passive dynamics posited in Eq. (15). A simple calculation gives:

$$\mathbf{dx}_t^c = [-\mathbf{F} + 2\mathbf{D}\nabla \ln \rho(\mathbf{x}_t)]dt + \sqrt{2\mathbf{D}}d\mathbf{W}_t^c + \Delta\mathbf{x}dJ \left(\frac{\lambda(-\Delta\mathbf{x})\rho(\mathbf{x}_t + \Delta\mathbf{x})}{\rho(\mathbf{x}_t)} \right). \quad (\text{C15})$$

Thus given a steady-state distribution $\rho(\mathbf{x})$, probabilistic time reversal gives an explicit optimal control protocol for state-independent passive dynamics.

We conjecture, and show in the main text through examples, that probabilistically time-reversing the passive dynamics yields a near-optimal protocol more generally, approaching the bound (12) in the high-information limit. Here we derive an analytic upper bound on the information rate required to implement the time-reversal protocol, allowing quantitative estimation of the protocol's distance from the bound. For arbitrary passive dynamics as given by Eq. (1), the time-reversal protocol is

$$\mathbf{dx}_t^c = [-\mathbf{F}(\mathbf{x}_t) + 2\mathbf{D}\nabla \ln \rho(\mathbf{x}_t)]dt + \sqrt{2\mathbf{D}}d\mathbf{W}_t^c + \Delta\mathbf{x}dJ \left(\frac{\lambda(\mathbf{x}_t + \Delta\mathbf{x}, -\Delta\mathbf{x})\rho(\mathbf{x}_t + \Delta\mathbf{x})}{\rho(\mathbf{x}_t)} \right). \quad (\text{C16})$$

This protocol depends only on the current state $\mathbf{x}_t$, so the instantaneous control information rate $\dot{I}_c^+$ is an upper bound on the full-history transfer entropy rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{dx}^c}$ (Eq. (A4)). Thus, the difference between the information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{dx}^c}$ and the bound $\dot{S}_p$ is upper-bounded by

$$\begin{aligned} \dot{\mathcal{T}}_{\mathbf{x} \rightarrow \mathbf{dx}^c} - \dot{S}_p &\leq \dot{I}_c^+ - \dot{S}_p \\ &= \frac{1}{4} \left[\langle \mathbf{F} \cdot \mathbf{D}^{-1} \mathbf{F} \rangle_\rho - \langle \mathbf{F} \rangle_\rho \cdot \mathbf{D}^{-1} \langle \mathbf{F} \rangle_\rho \right] + \iint d\mathbf{x}d\Delta\mathbf{x} \rho(\mathbf{x}) \lambda(\mathbf{x}, \Delta\mathbf{x}) \ln \frac{\lambda(\mathbf{x}, \Delta\mathbf{x})}{\bar{\lambda}(\Delta\mathbf{x})}, \end{aligned} \quad (\text{C17})$$

where we used Eqs. (5) and (A7), and defined $\bar{\lambda}(\Delta\mathbf{x}) \equiv \langle \lambda(\mathbf{x}, \Delta\mathbf{x}) \rangle_\rho$.

Appendix D: Controlling a Brownian Particle in a Harmonic Trap

As an additional example beyond those covered in the main text, we consider the problem of controlling a Brownian particle in a harmonic trap, with passive dynamics (in dimensionless units)

$$d\mathbf{x}_t^p = -\mathbf{x}_t dt + \sqrt{2}d\mathbf{W}_t^p. \quad (\text{D1})$$

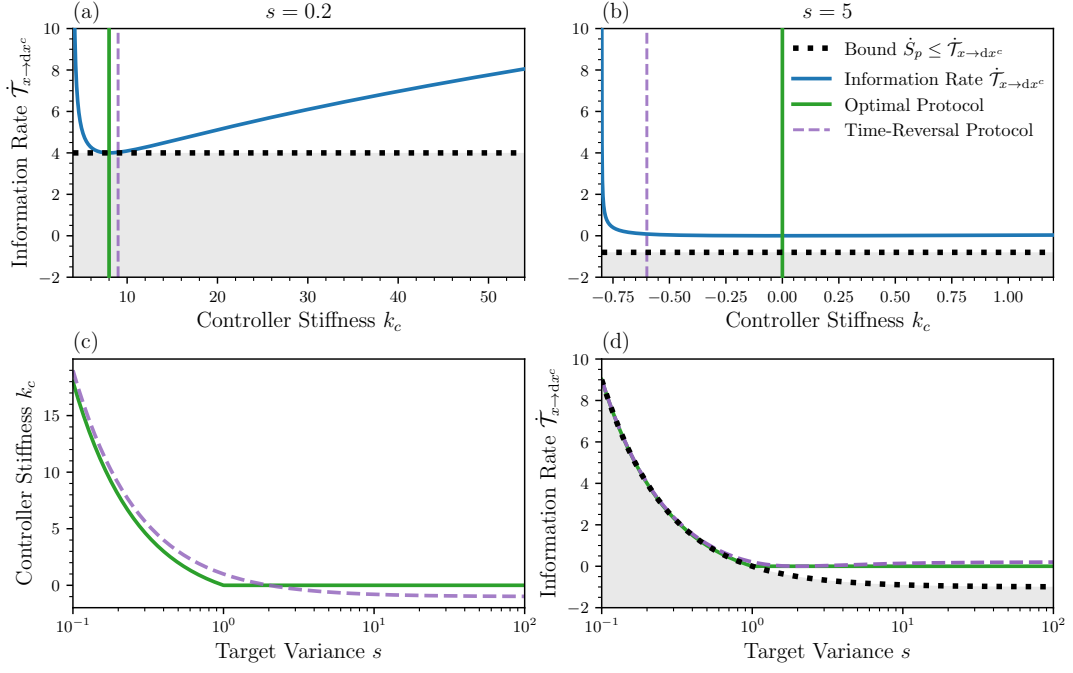


FIG. 6. Control of a Brownian particle in a harmonic trap. (a) and (b) Information rate (blue) as a function of controller stiffness k_c for target variance $s = 0.2$ (a) and $s = 5$ (b). Black dotted line: bound, Eq. (12). Vertical lines: optimal protocol (green) and time-reversal protocol (purple). (c) Controller stiffness as a function of target variance s for the optimal (green) and time-reversal (purple) protocols. (d) Information rate vs target variance. Black: bound, Eq. (12). Green: optimal protocol. Purple: time-reversal protocol.

Here the goal of the controller is to achieve a target variance $s = \text{Var}(x)$ using the minimum required information rate $\dot{\mathcal{T}}_{x \rightarrow dx^c}$. We stipulate that the controller apply a noisy linear force, as

$$dx_t^c = -k_c x_t dt + \sqrt{2D_c} dW_t^c. \quad (\text{D2})$$

For fixed s , the control protocol has only one degree of freedom, which we take to be k_c . The controller noise is then $D_c = s(1 + k_c) - 1$.

The passive entropy rate is $\dot{S}_p = 1/s - 1$, which in turn gives a lower bound on the information rate from Eq. (12). Intriguingly, we see that when $s > 1$, i.e., the target variance is larger than the equilibrium variance of the passive dynamics alone, the passive entropy rate becomes negative. Since the information rate must be nonnegative, for $s > 1$ the bound is thus no longer tight. We can nonetheless compute the information rate,

$$\dot{\mathcal{T}}_{x \rightarrow dx^c} = \frac{1}{2} \left[\sqrt{1 + \frac{k_c^2}{s(1 + k_c) - 1}} - 1 \right], \quad (\text{D3})$$

and minimize it with respect to the control stiffness k_c to obtain an optimal protocol given by $k_c = 2 \max\{0, 1/s - 1\}$. The time-reversal protocol, meanwhile, is $k_c = 2/s - 1$. While these protocols are not equal, the time-reversal protocol is nonetheless near-optimal, and requires the same information rate in the high-information (or equivalently low- s) limit [Fig. 6].

Appendix E: Details for Bit Flip Control

Here we provide expanded details and derivations for the bit flip control example in the main text. This application consists of controlling a single bit (with $x \in \{0, 1\}$), which passively flips its state at rate γ , to maximize the probability ρ_0 of it being in state 0. From Eq. (5), the passive entropy rate is

$$\dot{S}_p = (2\rho_0 - 1)\gamma \ln \frac{\rho_0}{1 - \rho_0}. \quad (\text{E1})$$

This straightforwardly gives the Pareto frontier between the performance metric $\mathcal{P}[\rho] = \rho_0$ and the information rate,

$$\dot{\mathcal{T}}_{x \rightarrow dx^c} \geq (2\rho_0 - 1)\gamma \ln \frac{\rho_0}{1 - \rho_0}. \quad (\text{E2})$$

Given γ , the controller can choose the control jump rates λ_0 and λ_1 to achieve a target performance $\mathcal{P}[\rho] = \rho_0$ with limited information. To enforce a steady-state, the rates and probabilities must satisfy

$$\rho_0(\gamma + \lambda_0) = (1 - \rho_0)(\gamma + \lambda_1). \quad (\text{E3})$$

We can thus write λ_1 in terms of γ, ρ_0 , and λ_0 as

$$\lambda_1 = \frac{\rho_0}{1 - \rho_0}(\gamma + \lambda_0) - \gamma. \quad (\text{E4})$$

The optimal control protocol (17) which probabilistically time-reverses the passive dynamics is

$$\lambda_0 = \gamma \frac{1 - \rho_0}{\rho_0}. \quad (\text{E5})$$

To explore the information rates of suboptimal protocols, we compute the information rate $\dot{\mathcal{T}}_{x \rightarrow dx^c}$. To do this, we define the probability $q_t \equiv p(x_t = 1 | \{d\mathbf{x}^c\})$, which we can think of as a time-evolving belief about the current state of the bit. We also define the expected rate of controlled jumps given the controller history, $\ell_t(q) \equiv \mathbb{E}[\lambda_x | \{d\mathbf{x}^c\}] = (1 - q_t)\lambda_0 + q_t\lambda_1$. Using Eq. (A2), we have

$$\dot{\mathcal{T}}_{x \rightarrow dx^c} = \left\langle \lambda_x \ln \frac{\lambda_x}{\ell(q)} \right\rangle_{p(q)}. \quad (\text{E6})$$

Evaluating this average requires computing the steady-state average $p(q)$, which we will do by constructing a differential equation describing the dynamics of q_t . First consider the time between discrete controller actions. When no controller action takes place, the passive dynamics push the belief q_t towards $1/2$ at rate

$$\dot{q} = \gamma(1 - 2q). \quad (\text{E7})$$

The absence of a controller action is also informative. Since $\lambda_1 \geq \lambda_0$ (as required for $\rho_0 \geq 1/2$), this absence of a control action updates the belief towards $x = 0$ at rate

$$\dot{q} = -(\lambda_1 - \lambda_0)q(1 - q). \quad (\text{E8})$$

Finally, controlled jumps in state provide discrete updates to the belief. At the instantaneous moment of a controlled jump, the post-jump belief $p(x_t^+ | \text{jump at } t)$ must satisfy

$$q_t^+ = p(x_t^+ | \text{controlled jump at } t) = \frac{p(\text{controlled jump at } t | x_t^+)p(x_t^+)}{p(\text{controlled jump at } t)} = \frac{\lambda_0(1 - q_t^-)}{\ell_t(q_t^-)} \equiv R(q_t^-). \quad (\text{E9})$$

Here the notation $q_t^+ = R(q_t^-)$ means that q_t^+ is the post-jump belief given a controlled jump at time t and pre-jump belief q_t^- . Combining all of this, we obtain the following Fokker–Planck dynamics for $p(q)$:

$$\partial_t p_t(q) = -\partial_q \left[\underbrace{(\gamma(1 - 2q) - (\lambda_1 - \lambda_0)q(1 - q))}_{f(q)} p_t(q) \right] - \ell(q)p_t(q) + \int dy \ell(y)p_t(y)\delta(q - R(y)). \quad (\text{E10})$$

We can simplify this significantly by noting that the function R is its own inverse, $R(R(q)) = q$, and using the delta function identity $\delta(q - R(y)) = \delta(y - R(q))/|R'(R(q))|$, to evaluate the integral

$$\int dy \ell(y)p_t(y)\delta(q - R(y)) = \frac{(\lambda_0\lambda_1)^2}{\ell(q)^3} p_t(R(q)). \quad (\text{E11})$$

Thus the final equation defining the steady-state belief probability $p(q)$ is

$$\partial_q [f(q)p(q)] + \ell(q)p(q) = \frac{(\lambda_0\lambda_1)^2}{\ell(q)^3} p(R(q)). \quad (\text{E12})$$

Solving numerically for $p(q)$ is then sufficient to calculate the information rate using Eq. (E6).

Appendix F: Details for Mexican-hat Potential

Here we provide details for calculating the Pareto frontiers and optimal control protocols for the 2D Brownian particle in a Mexican-hat potential. From Eq. (5), the passive entropy rate is

$$\dot{S}_p = \mathcal{I}[\rho] + 2 - 4\langle r^2 \rangle_\rho; \quad (\text{F1})$$

this in turn sets the general lower bound on the information rate required for control. We will consider the control problem of minimizing the mean-squared distance from two different fixed points: $\mathbf{x}_1^* = (0,0)$, and $\mathbf{x}_2^* = (1,0)$. We will determine Pareto frontiers using the Schrödinger equation mapping from App. B. The Schrödinger equation corresponding to the control performance extremization, applied to this system, is

$$[-4\nabla^2 + 2 - 4r^2 - \lambda|\mathbf{x} - \mathbf{x}^*|^2] \psi(\mathbf{x}) = E_0(\lambda)\psi(\mathbf{x}). \quad (\text{F2})$$

For mathematical convenience, we consider minimization of $\langle |\mathbf{x} - \mathbf{x}^*|^2 \rangle_\rho$ rather than maximization of its inverse.

We begin with the first target $\mathbf{x}_1^*$. The Schrödinger equation is then

$$[-4\nabla^2 + 2 - (4 + \lambda)r^2] \psi(\mathbf{x}) = E_0(\lambda)\psi(\mathbf{x}). \quad (\text{F3})$$

This is a standard isotropic 2D quantum harmonic oscillator, and thus has ground-state energy

$$E_0(\lambda) = 2 + 4\sqrt{-\lambda - 4}. \quad (\text{F4})$$

This requires $\lambda \leq -4$. Using the formula for the Pareto frontier (B8), we obtain

$$\langle |\mathbf{x} - \mathbf{x}^*|^2 \rangle_\rho^{-1} \leq \frac{1}{8} \left(\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \text{d}\mathbf{x}^c} - 2 + \sqrt{\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \text{d}\mathbf{x}^c}^2 - 4\dot{\mathcal{T}}_{\mathbf{x} \rightarrow \text{d}\mathbf{x}^c} + 68} \right). \quad (\text{F5})$$

The ground-state probability density, meanwhile, is

$$\rho(\mathbf{x}) = \frac{\sqrt{-\lambda - 4}}{2\pi} \exp \left(-\frac{\sqrt{-\lambda - 4}}{2} [\mathbf{x} - \mathbf{x}_1^*]^2 \right). \quad (\text{F6})$$

We then consider the second target $\mathbf{x}_2^*$. The Schrödinger equation is

$$[-4\nabla^2 + 2 - 4r^2 - \lambda|\mathbf{x} - \mathbf{x}_2^*|^2] \psi(\mathbf{x}) = E_0(\lambda)\psi(\mathbf{x}). \quad (\text{F7})$$

Using $\mathbf{x} = (x, y)$ and completing the square, we can rewrite this as

$$\begin{aligned} E_0(\lambda)\psi(\mathbf{x}) &= [-4\nabla^2 + 2 - 4(x^2 + y^2) - \lambda((x-1)^2 + y^2)] \psi(\mathbf{x}) \\ &= \left[-4\nabla^2 + 2 - (4 + \lambda) \left[\left(x - \frac{\lambda}{4 + \lambda} \right)^2 + y^2 \right] - \lambda + \frac{\lambda^2}{4 + \lambda} \right] \psi(\mathbf{x}). \end{aligned} \quad (\text{F8})$$

The ground-state energy is then

$$E_0(\lambda) = 2 + \frac{\lambda^2}{4 + \lambda} - \lambda + 4\sqrt{-\lambda - 4}, \quad (\text{F9})$$

while the probability density is

$$\rho(\mathbf{x}) = \frac{\sqrt{-\lambda - 4}}{2\pi} \exp \left(-\frac{\sqrt{-\lambda - 4}}{2} \left[\left(x - \frac{\lambda}{4 + \lambda} \right)^2 + y^2 \right] \right). \quad (\text{F10})$$

Here we see the distribution is offset slightly from the target $x = 1$, approaching it only asymptotically as $\lambda \rightarrow -\infty$.

Appendix G: Details for Microbial Navigation

1. Optimal Control Protocol with Full Observability

For the navigation problem, the corresponding Schrödinger equation is

$$[-4D_r\partial_\theta^2 - \lambda \cos \theta] \psi(\theta) = E_0(\lambda) \psi(\theta). \quad (\text{G1})$$

This is the Hamiltonian for a quantum rotor which has been studied previously [51]. The ground-state distribution is

$$\rho(\theta) = \frac{1}{2\pi} \text{ce}_0(\theta/2, -\lambda/2D_r)^2, \quad (\text{G2})$$

with corresponding ground-state energy $a_0(-\lambda/2D_r)$, where ce_0 is the zeroth order Mathieu C function of the first kind [52], and a_0 is the zeroth order Mathieu characteristic function.

From Eq. (17), the optimal control force is

$$F_c(\theta) = 4D_r \frac{\partial_\theta \text{ce}_0\left(\theta/2, b_0^{-1}\left(\dot{\mathcal{T}}_{\theta \rightarrow d\theta^c}/D_r\right)\right)}{\text{ce}_0\left(\theta/2, b_0^{-1}\left(\dot{\mathcal{T}}_{\theta \rightarrow d\theta^c}/D_r\right)\right)}. \quad (\text{G3})$$

2. Run-and-Tumble Navigation

Now consider a Run-and-Tumble strategy, where the only action available is to fully reorient the heading with rate $\lambda(\{\theta\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^t)$ such that the angle change is uniformly distributed on $[0, 2\pi]$. We consider the Markovian version of this strategy where $\lambda = \lambda(\theta)$.

The transfer entropy rate can be computed exactly for this strategy. To do so, we note that the entire history of the state and controller are fully characterized by the current heading θ and the time τ since the last tumble. Because the passive dynamics are Markovian and the tumbles fully reset the heading, no other history carries further information. Thus we need only determine the distribution $q(\theta, \tau)$, which is the probability that a run still survives at time τ and the current heading is θ . Integrating over θ and τ does not equal one, so this is not a normalized joint probability distribution. Integrating over θ gives the survival function

$$S(\tau) = \int d\theta q(\theta, \tau), \quad (\text{G4})$$

which is the probability of a run having survived without interruption at time τ . Integrating this over τ in turn gives the mean run time,

$$\bar{\tau} = \int_0^\infty S(\tau) d\tau. \quad (\text{G5})$$

The normalized probability distribution over run time τ and heading θ is $p(\theta, \tau) = q(\theta, \tau)/\bar{\tau}$. The distribution $q(\theta, \tau)$ evolves according to

$$\partial_\tau q(\theta, \tau) = D_r \partial_\theta^2 q(\theta, \tau) - \lambda(\theta) q(\theta, \tau), \quad (\text{G6})$$

with initial condition $q(\theta, 0) = 1/2\pi$. This makes it clear it is not a normalized joint probability. This PDE is linear and thus easily solvable numerically given $\lambda(\theta)$.

The conditional distribution over the heading given a time τ since the last tumble is $\rho(\theta|\tau) = q(\theta, \tau)/S(\tau)$. We then define $\bar{\lambda}(\tau) = \int d\theta \rho(\theta|\tau) \lambda(\theta)$ as the mean tumble rate given time τ since the last tumble. It should also satisfy

$$d_\tau S(\tau) = -\bar{\lambda}(\tau) S(\tau), \quad (\text{G7})$$

with $S(0) = 1$.

The transfer entropy rate, given that the control strategy is Markovian, is computed as

$$\begin{aligned} \dot{\mathcal{T}}_{\theta \rightarrow d\theta^c} &= \left\langle \lambda(\theta) \ln \frac{\lambda(\theta)}{\mathbb{E}[\lambda(\theta) | \{d\theta^c\}_{-\infty}^t]} \right\rangle_{p(\theta, \{d\theta^c\}_{-\infty}^t)} \\ &= \left\langle \lambda(\theta) \ln \frac{\lambda(\theta)}{\bar{\lambda}(\tau)} \right\rangle_{p(\theta, \tau)}. \end{aligned} \quad (\text{G8})$$

3. Optimal Protocol with Limited Observability

Run-and-Tumble is not the optimal control protocol using only measurements of $\cos \theta$ in the low-information limit. Here we prove that Run-Reverse is the unique optimal control protocol in the low-information limit, even allowing the use of arbitrary memory, given that only $\cos \theta$ can be measured. For mathematical simplicity, we will consider control using only measurements of $|\theta|$ which is equivalent to using only $\cos \theta$.

First, note that as shown above, maximizing up-gradient velocity $v/v_0 = \langle \cos \theta \rangle$ given fixed passive entropy rate $\dot{S}_p$ requires a steady-state distribution

$$\rho(\theta) = \frac{1}{2\pi} c_{e0} \left[\theta/2, b_0^{-1} \left(\dot{\mathcal{T}}_{\theta \rightarrow d\theta^c} / D_r \right) \right]^2. \quad (\text{G9})$$

To saturate the bound, we must then find a control protocol that achieves this distribution using only measurements of $|\theta|$, while also saturating the bound (12).

Expanding $\rho(\theta)$ in the limit of small information rate, $\epsilon \equiv \left(\dot{\mathcal{T}}_{\theta \rightarrow d\theta^c} / D_r \right)^{1/2} \ll 1$, gives

$$\rho(\theta) = \frac{1}{2\pi} \left[1 + \sqrt{2} \epsilon \cos \theta \right] + \mathcal{O}(\epsilon^2). \quad (\text{G10})$$

In this same limit, the passive entropy rate is

$$\dot{S}_p = \epsilon^2 D_r + \mathcal{O}(\epsilon^3). \quad (\text{G11})$$

We similarly expand the control protocol around zero information as

$$\lambda(\Delta\theta | \{|\theta|\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt}) = r(\Delta\theta | \{d\theta^c\}_{-\infty}^{t-dt}) + \epsilon \eta(\Delta\theta | \{|\theta|\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt}) + \mathcal{O}(\epsilon^2). \quad (\text{G12})$$

Here η and r are functions which do and do not respectively depend on the state θ and its history. Note that using arbitrary $\lambda(\Delta\theta, |\theta|)$ can describe any arbitrary additive jump or drift-diffusion control protocol [19]. When $\epsilon = 0$ this protocol has zero information rate.

Now consider the gap in the bound (12), which we showed in App. C can be written as

$$\Delta \mathcal{T}_{\theta \rightarrow d\theta^c} - \Delta S_p = \underbrace{I[d\theta_t^c; \{\theta\}_{-\infty}^{t-dt} | \theta_t, \{d\theta^c\}_{-\infty}^{t-dt}]}_{\text{Term 1}} + \underbrace{I[y_t; \{d\theta^c\}_{-\infty}^t | \theta_{t+dt}]}_{\text{Term 2}}. \quad (\text{G13})$$

Since $\dot{S}_p \sim \epsilon^2$ to leading order, saturating the bound in the low-information limit requires that both term 1 and term 2 scale as ϵ^3 .

We consider term 1 explicitly. Defining $\bar{\eta}(\Delta\theta, |\theta|_t, \{d\theta^c\}_{-\infty}^{t-dt}) = \langle \eta(\Delta\theta, \{|\theta|\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt}) | |\theta|_t, \{d\theta^c\}_{-\infty}^{t-dt} \rangle$, the leading-order term is

$$\frac{1}{dt} I[d\theta_t^c; \{\theta\}_{-\infty}^{t-dt} | \theta_t, \{d\theta^c\}_{-\infty}^{t-dt}] = \frac{\epsilon^2}{2} \left\langle \int d\Delta\theta \frac{[\eta(\Delta\theta, \{|\theta|\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt}) - \bar{\eta}(\Delta\theta, |\theta|_t, \{d\theta^c\}_{-\infty}^{t-dt})]^2}{r(\Delta\theta, \{d\theta^c\}_{-\infty}^{t-dt})} \right\rangle + \mathcal{O}(\epsilon^3). \quad (\text{G14})$$

In this section, we use angle brackets to denote averages over the full distribution $p(\{\theta\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt})$. The leading-order term is zero if and only if $\eta = \bar{\eta}$, so that the optimal control strategy cannot depend on memory of past states in the low-information limit.

We then note that since the optimal distribution in the small- ϵ limit differs from uniform only in the first Fourier mode, the control protocol should likewise depend only on the first Fourier mode to first order. Any dependence on higher Fourier modes will increase the information rate without improving performance. While dependence on both $\sin \theta$ and $\cos \theta$ can both be used to control the first Fourier mode, only $\cos \theta$ satisfies the constraint that only $|\theta|$ can be used. Thus we can write

$$\lambda(\Delta\theta, |\theta|, \{d\theta^c\}_{-\infty}^{t-dt}) = r(\Delta\theta | \{d\theta^c\}_{-\infty}^{t-dt}) + \epsilon A(\Delta\theta, \{d\theta^c\}_{-\infty}^{t-dt}) \cos \theta + \mathcal{O}(\epsilon^2). \quad (\text{G15})$$

We now impose stationarity of the steady-state distribution, or equivalently of the first Fourier mode in the small- ϵ limit. Passive diffusion decreases the mode $\cos \theta$ at a rate $\epsilon D_r / \sqrt{2}$, so the active control must increase the first mode at the same rate. A jump by $\Delta\theta$ increases the first mode $\cos \theta$ by

$$\begin{aligned} \Delta \cos \theta &= \cos(\theta + \Delta\theta) - \cos \theta \\ &= -[1 - \cos \Delta\theta] \cos \theta - \sin \theta \sin \Delta\theta \\ &= -z(\Delta\theta) \cos \theta - \sin \theta \sin \Delta\theta, \end{aligned} \quad (\text{G16})$$

where we have defined the function $z(\Delta\theta) \equiv 1 - \cos \Delta\theta$. The active control therefore increases the first mode at a rate

$$\begin{aligned} & \left\langle \int d\Delta\theta \lambda(\Delta\theta, |\theta|, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) [-z(\Delta\theta) \cos \theta - \sin \theta \sin \Delta\theta] \right\rangle \\ &= -\frac{\epsilon}{\sqrt{2}} \left\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \right\rangle - \frac{\epsilon}{2} \left\langle \int d\Delta\theta A(\Delta\theta, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \right\rangle + \mathcal{O}(\epsilon^2) \\ &= +\frac{\epsilon D_r}{\sqrt{2}} + \mathcal{O}(\epsilon^2). \end{aligned} \quad (\text{G17})$$

Here in the last line we used the fact that to first order in ϵ this must be equal to the passive diffusion rate $-\epsilon D_r/\sqrt{2}$. We can now directly evaluate the transfer entropy rate, which to second order in ϵ is

$$\begin{aligned} \dot{\mathcal{T}}_{\theta \rightarrow \mathrm{d}\theta^c} &= \frac{\epsilon^2}{2} \left\langle \int d\Delta\theta \frac{[A(\Delta\theta, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) \cos \theta]^2}{r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t})} \right\rangle + \mathcal{O}(\epsilon^3) \\ &= \frac{\epsilon^2}{4} \left\langle \int d\Delta\theta \frac{A(\Delta\theta, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t})^2}{r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t})} \right\rangle + \mathcal{O}(\epsilon^3). \end{aligned} \quad (\text{G18})$$

We then apply the Cauchy-Schwarz inequality, taking $X = A/\sqrt{r}$ and $Y = \sqrt{r}z$ such that $\langle X^2 \rangle \geq |\langle XY \rangle|^2 / \langle Y^2 \rangle$. Combined with Eq. (G17), this gives

$$\begin{aligned} \dot{\mathcal{T}}_{\theta \rightarrow \mathrm{d}\theta^c} &\geq \frac{\epsilon^2}{4} \frac{\langle \int d\Delta\theta A(\Delta\theta, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle^2}{\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta)^2 \rangle} + \mathcal{O}(\epsilon^3) \\ &= \frac{\epsilon^2}{2} \frac{(\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle + D_r)^2}{\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta)^2 \rangle} + \mathcal{O}(\epsilon^3). \end{aligned} \quad (\text{G19})$$

Recalling that $z(\Delta\theta) = 1 - \cos \Delta\theta$, we note that $z(\Delta\theta) \leq 2$, so that we can write

$$\left\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta)^2 \right\rangle \leq 2 \left\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \right\rangle. \quad (\text{G20})$$

Using this inequality in Eq. (G19) then gives

$$\begin{aligned} \dot{\mathcal{T}}_{\theta \rightarrow \mathrm{d}\theta^c} &\geq \frac{\epsilon^2}{4} \frac{(\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle + D_r)^2}{\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle} + \mathcal{O}(\epsilon^3) \\ &= \epsilon^2 D_r + \frac{\epsilon^2}{4} \frac{(\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle - D_r)^2}{\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle} + \mathcal{O}(\epsilon^3). \end{aligned} \quad (\text{G21})$$

Consistency with our definition $\epsilon \equiv (\dot{\mathcal{T}}_{\theta \rightarrow \mathrm{d}\theta^c}/D_r)^{1/2} \ll 1$ requires $\dot{\mathcal{T}}_{\theta \rightarrow \mathrm{d}\theta^c} = \epsilon^2 D_r$ to second order. Thus saturation of the Pareto frontier in the low-information limit requires both saturating the inequality (G21), and

$$\frac{(\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle - D_r)^2}{\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \rangle} = 0. \quad (\text{G22})$$

The inequality, Eq. (G21), came from applying both the Cauchy-Schwarz inequality as well as $z(\Delta\theta) \leq 2$. For the second inequality, saturation is only possible for $\Delta\theta = \pi$. Saturating Cauchy-Schwarz requires $A/\sqrt{r} \propto \sqrt{r}z$, which with $z(\pi) = 2$ implies $A(\Delta\theta, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) \propto r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t})$. The remaining condition, Eq. (G22), requires

$$\left\langle \int d\Delta\theta r(\Delta\theta | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) z(\Delta\theta) \right\rangle = D_r. \quad (\text{G23})$$

With only jumps $\Delta\theta = \pi$ and $z(\pi) = 2$, this gives $\langle r(\pi | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) \rangle = D_r/2$. Eq. (G17) then implies $A(\pi, \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t}) = -2\sqrt{2}r(\pi | \{\mathrm{d}\theta^c\}_{-\infty}^{t-\mathrm{d}t})$.

Combining these results, the unique control protocol that saturates the Pareto frontier to first order in ϵ is

$$\lambda(\Delta\theta|\{|\theta|\}_{-\infty}^t, \{d\theta^c\}_{-\infty}^{t-dt}) = r(\pi|\{d\theta^c\}_{-\infty}^{t-dt}) \left[1 - \epsilon 2\sqrt{2} \cos\theta\right] + \mathcal{O}(\epsilon^2). \quad (\text{G24})$$

This proves that, subject to the constraint of only observing $|\theta|$, the optimal protocol in the low-information limit uses no memory of past observations, and only induces discrete jumps by $\Delta\theta = \pi$. This is a classical Run-Reverse strategy.

The remaining degeneracy allows for arbitrary dependence of the jump rate on the control history $\{d\theta^c\}_{-\infty}^{t-dt}$. A specific choice with no control-history dependence would be $r(\pi|\{d\theta^c\}_{-\infty}^{t-dt}) = \langle r(\pi|\{d\theta^c\}_{-\infty}^{t-dt}) \rangle$, which leads to the control protocol

$$\lambda(\Delta\theta|\{|\theta|\}_{-\infty}^t) = \frac{D_r}{2} \left[1 - \epsilon 2\sqrt{2} \cos\theta\right] \delta(\Delta\theta - \pi) + \mathcal{O}(\epsilon^2), \quad (\text{G25})$$

which exactly recovers the optimal reverse rate function derived in Ref. [19].

Appendix H: Details for Information Engine

1. Deriving the Pareto Frontier

Here we derive the Pareto frontier for the information engine, Eq. (30). Applying the bound (12) to the information engine, and evaluating the passive entropy rate, gives (using $u = z - x$, and the steady-state probability $\rho(u)$):

$$D\mathcal{I}[\rho] - \frac{\kappa}{\gamma} \leq \dot{\mathcal{T}}_{u \rightarrow dz^c}. \quad (\text{H1})$$

As in the main text, we define the timescale $\tau_r = 1/\beta D\kappa$ and dimensionless mass $\delta_g = mg/\kappa\sqrt{D\tau_r}$, as well as the dimensionless length scale $\sigma = \sqrt{D\tau_r}$. We can write the steady-state upward velocity of the particle as

$$v\tau_r/\sigma = \langle u \rangle/\sigma - \delta_g. \quad (\text{H2})$$

To do no work on the particle on average, the controller must not change the potential energy $V(x, z) = \frac{1}{2}\kappa(x - z)^2$. Equivalently, the control updates must not change $\langle u^2 \rangle$. It follows that at steady state the passive contribution to the rate of change of $\langle u^2 \rangle$ must also vanish. This results in a one-to-one relationship between the variance of u and the velocity: $\text{Var}(u)/\sigma^2 = 1 - \delta_g\nu - \nu^2$, where we have defined the dimensionless velocity $\nu \equiv v\tau_r/\sigma$.

The Cramér–Rao inequality gives $\mathcal{I}[\rho] \geq 1/\text{Var}(u)$, which, combined with the bound (H1), gives

$$\frac{1}{1 - \delta_g\nu - \nu^2} - 1 \leq \tau_r \dot{\mathcal{T}}_{u \rightarrow dz^c}. \quad (\text{H3})$$

Solving for ν yields a velocity–information frontier:

$$\nu \leq \frac{-\delta_g + \sqrt{\delta_g^2 + 4 \left(1 + \left(\tau_r \dot{\mathcal{T}}_{u \rightarrow dz^c}\right)^{-1}\right)}}{2}. \quad (\text{H4})$$

The velocity limit increases monotonically with information, asymptotically approaching the finite upper bound $\frac{1}{2} \left(\sqrt{\delta_g^2 + 4} - \delta_g\right)$ as $\dot{\mathcal{T}}_{u \rightarrow dz^c} \rightarrow \infty$. In this way, achieving the maximum performance of the engine requires a diverging information rate. Multiplying by δ_g leads to the Pareto frontier in the main text between extracted heat and information rate, Eq. (30). Optimizing over δ_g gives a more general bound,

$$\tau_r \beta \dot{Q} \leq \frac{1}{1 + \left(\tau_r \dot{\mathcal{T}}_{u \rightarrow dz^c}\right)^{-1}}. \quad (\text{H5})$$

2. Second Law for the Information Engine

Here we show that the second law implies $\beta\dot{Q} \leq \dot{\mathcal{T}}_{u \rightarrow dz^c}$. Let x and z be bipartite degrees of freedom, as described in Ref. [29]. Then the second law of bipartite thermodynamics applied to the x subsystem gives [27, 29]

$$\beta\dot{Q} \leq \dot{I}_Z \equiv \lim_{dt \rightarrow 0} \frac{I[x_t; z_{t+dt}] - I[x_t; z_t]}{dt}, \quad (\text{H6})$$

for information flow $\dot{I}_Z$ defined as the instantaneous rate at which the z dynamics increase the mutual information between x and z . This is a lower bound on the transfer entropy rate from x to z , giving [28]

$$\dot{I}_Z \leq \dot{\mathcal{T}}_{x \rightarrow z} \equiv \lim_{dt \rightarrow 0} \frac{1}{dt} I[z_{t+dt}; \{x\}_{-\infty}^t | \{z\}_{-\infty}^t]. \quad (\text{H7})$$

Finally, for the particular information engine setup, the transfer entropy rate $\dot{\mathcal{T}}_{x \rightarrow z}$ is equal to the control information rate $\dot{\mathcal{T}}_{u \rightarrow dz^c}$. To see this, note that the dynamics depend only on the state $u = z - x$, not the absolute values of z or x , so that

$$\begin{aligned} \dot{\mathcal{T}}_{x \rightarrow z} &\equiv \lim_{dt \rightarrow 0} \frac{1}{dt} I[z_{t+dt}; \{x\}_{-\infty}^t | \{z\}_{-\infty}^t] \\ &= \lim_{dt \rightarrow 0} \frac{1}{dt} I[dz_t^c; \{x\}_{-\infty}^t | \{z\}_{-\infty}^t] \\ &= \lim_{dt \rightarrow 0} \frac{1}{dt} I[dz_t^c; \{u\}_{-\infty}^t | \{z\}_{-\infty}^t] \\ &= \lim_{dt \rightarrow 0} \frac{1}{dt} I[dz_t^c; \{u\}_{-\infty}^t | \{dz^c\}_{-\infty}^{t-dt}] \\ &= \dot{\mathcal{T}}_{u \rightarrow dz^c}. \end{aligned} \quad (\text{H8})$$

Thus the second law implies $\beta\dot{Q} \leq \dot{\mathcal{T}}_{u \rightarrow dz^c}$.

3. Experimentally Implemented Control Protocol

As described in the main text, the controller makes noisy measurements of the particle position $\hat{x}_t = x_t + \mathcal{N}(0, \sigma_m^2)$ (with measurement noise σ_m) at periodic time intervals of length t_s . The protocol implemented in Ref. [43] updates the trap center based only on the measured particle position as

$$dz_{t=nt_s}^c = \begin{cases} \alpha(\hat{x}_t - z_t), & \hat{x}_t > z_t, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{H9})$$

for integer n , and feedback gain parameter α that is selected to ensure the controller does no average work on the particle.

4. Probabilistic Time-Reversal Protocol

Here we construct the time-reversal control protocol for the information engine shown in Fig. 5. We will initially neglect the measurement noise, then restore it at the end of this subsection.

For notational simplicity, we use the dimensionless variable $\xi = (z - x)/\sigma$, which obeys passive dynamics

$$d\xi_\tau = (\delta_g - \xi)d\tau + \sqrt{2}dW_\tau^p + d\xi_\tau^c. \quad (\text{H10})$$

Here $\tau = t/\tau_r$ is scaled time. The controller acts at discrete time intervals of scaled time $\Delta = t_s/\tau_r$. We will use ξ^- and ξ^+ for the states immediately before and after a control update respectively. The control action is thus $d\xi^c = \xi^+ - \xi^- = dz^c/\sigma$.

Saturating the Pareto frontier requires a Gaussian distribution (to saturate the Cramér–Rao inequality), so we take ξ^- and ξ^+ to be Gaussian-distributed with means $m_\pm$ and variances $s_\pm$, as

$$\xi^- \sim \mathcal{N}(m_-, s_-), \quad (\text{H11a})$$

$$\xi^+ \sim \mathcal{N}(m_+, s_+). \quad (\text{H11b})$$

The mean velocity of the particle in scaled units is thus $\nu = (m_+ - m_-)/\Delta$. Now consider the passive kernel from ξ^+ to ξ^- over time Δ ,

$$K_p(\xi^-|\xi^+) = \mathcal{N}[\xi^+ + (\delta_g - \xi^+)(1 - e^{-\Delta}), 1 - e^{-2\Delta}]. \quad (\text{H12})$$

This implies

$$m_- = m_+ + (\delta_g - m_+)(1 - e^{-\Delta}), \quad (\text{H13a})$$

$$s_- = s_+ e^{-2\Delta} + 1 - e^{-2\Delta}. \quad (\text{H13b})$$

Requiring that the trap does no average work on the particle enforces $\langle(\xi^-)^2\rangle = \langle(\xi^+)^2\rangle$, which in turn implies

$$m_-^2 + s_- = m_+^2 + s_+. \quad (\text{H14})$$

Thus we can write the two means and variances in terms of only the parameters Δ and δ_g , as well as the dimensionless velocity ν , as

$$m_+ = \delta_g + \frac{\Delta\nu}{1 - e^{-\Delta}}, \quad (\text{H15a})$$

$$m_- = \delta_g + \frac{\Delta\nu}{e^{\Delta} - 1}, \quad (\text{H15b})$$

$$s_+ = 1 - \left(\frac{\Delta\nu}{1 - e^{-\Delta}}\right)^2 - \frac{2\delta_g\Delta\nu}{1 - e^{-2\Delta}}, \quad (\text{H15c})$$

$$s_- = 1 - \left(\frac{\Delta\nu}{e^{\Delta} - 1}\right)^2 - \frac{2\delta_g\Delta\nu}{e^{2\Delta} - 1}. \quad (\text{H15d})$$

The reverse passive kernel, meanwhile, is

$$R_p(\xi^+|\xi^-) = \frac{p_+(\xi^+)K_p(\xi^-|\xi^+)}{p_-(\xi^-)} = \mathcal{N}\left[m^+ + \frac{s_+}{s_-}e^{-\Delta}(\xi^- - m_-), \frac{s_+}{s_-}(1 - e^{-2\Delta})\right]. \quad (\text{H16})$$

Thus to probabilistically time-reverse the passive dynamics, the control kernel must be

$$K_c(\xi^+|\xi^-) = R_p(\xi^+|\xi^-). \quad (\text{H17})$$

We can write the control, at time $\tau = n\Delta$ ($n \in \mathbb{N}$), in the differential notation used throughout the Article as

$$d\xi_\tau^c(\xi_\tau^-) = (m_+ - m_-) + \left(\frac{s_+}{s_-}e^{-\Delta} - 1\right)(\xi_\tau^- - m_-) + \sqrt{\frac{s_+}{s_-}(1 - e^{-2\Delta})}\mathcal{N}(0, 1). \quad (\text{H18})$$

We can now straightforwardly account for the measurement noise, by noting that the dimensionless pre-control measurement will be $\hat{\xi}_\tau^- = \xi_\tau^- + \mathcal{N}(0, \sigma_m^2/\sigma^2)$. We can then achieve the probabilistic time-reversal protocol, Eq. (H18), using the noisy measurement $\hat{\xi}_\tau^-$, by simply decreasing the noise added to the control update. The protocol, as a function of the noisy measurement, will be

$$d\xi_\tau^c(\hat{\xi}_\tau^-) = (m_+ - m_-) + \left(\frac{s_+}{s_-}e^{-\Delta} - 1\right)(\hat{\xi}_\tau^- - m_-) + \sqrt{\frac{s_+}{s_-}(1 - e^{-2\Delta}) - \left(\frac{s_+}{s_-}e^{-\Delta} - 1\right)^2 \frac{\sigma_m^2}{\sigma^2}}\mathcal{N}(0, 1), \quad (\text{H19})$$

which leads to a distribution $p(d\xi_\tau^c|\xi_\tau^-)$ identical to the target time-reversal control distribution with no measurement noise. This is achievable so long as the measurement noise satisfies the condition

$$\frac{s_+}{s_-}(1 - e^{-2\Delta}) - \left(\frac{s_+}{s_-}e^{-\Delta} - 1\right)^2 \frac{\sigma_m^2}{\sigma^2} \geq 0. \quad (\text{H20})$$

5. Information Rate for the Time-Reversal Protocol

For the time-reversal protocol, we can compute the transfer entropy rate semi-analytically. For a single controller update, the transfer entropy is

$$\begin{aligned}\dot{\mathcal{T}}_{\xi \rightarrow d\xi^c} &= \frac{1}{\Delta} I [d\xi_\tau^c; \{\xi\}_{-\infty}^\tau | \{d\xi^c\}_{-\infty}^{\tau-\Delta}] \\ &= \frac{1}{\Delta} I [d\xi_\tau^c; \xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta}] \\ &= \frac{1}{\Delta} (S [d\xi_\tau^c | \{d\xi^c\}_{-\infty}^{\tau-\Delta}] - S [d\xi_\tau^c | \xi_\tau^-, \{d\xi^c\}_{-\infty}^{\tau-\Delta}]).\end{aligned}\tag{H21}$$

Here in the second line we used the fact that the time-reversal control protocol depends only on the current pre-control state ξ_τ^- . The remaining task is then to evaluate the two conditional entropy terms. Given knowledge of ξ_τ^- the only remaining variability in the controller update is the Gaussian white noise, so the second conditional entropy is

$$S [d\xi_\tau^c | \xi_\tau^-, \{d\xi^c\}_{-\infty}^{\tau-\Delta}] = \frac{1}{2} \ln \left(2\pi e \frac{s_+}{s_-} (1 - e^{-2\Delta}) \right).\tag{H22}$$

The first conditional entropy also requires the variance $\text{Var}(d\xi_\tau^c | \{d\xi^c\}_{-\infty}^{\tau-\Delta})$. To compute this variance, we consider how it changes due to controlled and passive updates. Using Eq. (H18), the variance of the control update will be

$$\text{Var}(d\xi_\tau^c | \{d\xi^c\}_{-\infty}^{\tau-\Delta}) = \left(\frac{s_+}{s_-} e^{-\Delta} - 1 \right)^2 \text{Var}(\xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta}) + \frac{s_+}{s_-} (1 - e^{-2\Delta}).\tag{H23}$$

We can then write, noting that ξ_τ^- and $d\xi_\tau^c$ are jointly Gaussian distributed and using the standard equation for the conditional variance of a joint,

$$\text{Var}(\xi_\tau^- | d\xi_\tau^c, \{d\xi^c\}_{-\infty}^{\tau-\Delta}) = \text{Var}(\xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta}) - \frac{\text{Cov}(d\xi_\tau^c, \xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta})^2}{\text{Var}(d\xi_\tau^c | \{d\xi^c\}_{-\infty}^{\tau-\Delta})}.\tag{H24}$$

Here the conditional covariance evaluates to $\text{Cov}(d\xi_\tau^c, \xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta}) = \left(\frac{s_+}{s_-} e^{-\Delta} - 1 \right) \text{Var}(\xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta})$.

Since ξ_τ^+ is a deterministic translation of ξ_τ^- given $d\xi_\tau^c$, we also have $\text{Var}(\xi_\tau^+ | \{d\xi^c\}_{-\infty}^\tau) = \text{Var}(\xi_\tau^- | \{d\xi^c\}_{-\infty}^\tau)$. The passive update from time τ to $\tau + \Delta$ then leads to pre-control variance conditional on the control history at time $\tau + \Delta$ given by

$$\text{Var}(\xi_{\tau+\Delta}^- | \{d\xi^c\}_{-\infty}^\tau) = \text{Var}(\xi_\tau^+ | \{d\xi^c\}_{-\infty}^\tau) e^{-2\Delta} + 1 - e^{-2\Delta}.\tag{H25}$$

Using the fact that $\text{Var}(\xi_{\tau+\Delta}^- | \{d\xi^c\}_{-\infty}^\tau) = \text{Var}(\xi_\tau^- | \{d\xi^c\}_{-\infty}^{\tau-\Delta})$ at steady state, Eqs. (H23), (H24), and (H25) then constitute three coupled equations in three unknown variables. We then numerically solve for the variance $\text{Var}(d\xi_\tau^c | \{d\xi^c\}_{-\infty}^{\tau-\Delta})$ of the next control update given the control history in terms of only the parameters Δ , δ_g , and ν . The transfer entropy rate is then computed using Eq. (H21). Returning to physical units then simply requires dividing by the dimensional timescale τ_r .

Appendix I: Numerical Transfer Entropy Rate Calculations

When analytic calculations are not tractable, we numerically compute the information rate $\dot{\mathcal{T}}_{\mathbf{x} \rightarrow d\mathbf{x}^c}$ using the Path Weight Sampling method (TE-PWS) developed in Ref. [38]. This approach is used for the time-reversal protocol in Fig. 3, the Run-Reverse navigation strategy in Fig. 4, and the experimental information engine protocol in Fig. 5. Our code for implementing the TE-PWS method, adapted from code made public as part of Ref. [38], is available on GitHub [50].